



Review article

Decoding blood-brain barrier dysfunction in Alzheimer's Disease: Innovations and challenges in multimodal MRI and PET imaging biomarkers

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ABSTRACT

Alzheimer's disease (AD), a leading neurodegenerative disorder, involves blood-brain barrier (BBB) dysfunction as a critical contributor to its pathogenesis. This review synthesizes current advancements in in vivo magnetic resonance imaging (MRI) and positron emission tomography (PET) techniques for imaging BBB breakdown in AD. The BBB, a dynamic neurovascular interface, regulates amyloid-beta (A β) and tau clearance through specialized transporters and cellular interactions. BBB dysfunction, driven by tight junction disruption, transporter deficits, and pericyte degeneration, exacerbates A β accumulation and neuroinflammation. Dynamic contrast-enhanced MRI quantifies subtle leakage via gadolinium kinetics, while water-exchange MRI probes trans-BBB water dynamics without contrast agents. Dynamic glucose-enhanced MRI maps glucose transport anomalies linked to glucose transporter-1 dysfunction. PET imaging with tracers like [¹⁸F]-fluorodeoxyglucose and [¹¹C]-verapamil evaluates glucose metabolism and efflux transporter activity, revealing early metabolic deficits and impaired A β clearance. Challenges include low sensitivity for subtle leakage, model-dependent quantification, and spatial-temporal resolution trade-offs. Emerging strategies emphasize multimodal integration, ultrahigh-field systems, and artificial intelligence-driven analytics to decode region-specific BBB pathology. Longitudinal studies correlating imaging biomarkers with clinical progression and novel PET tracer development are pivotal for early diagnosis and personalized therapies. These innovations promise to elucidate BBB's role and promote a paradigm shift in diagnostic and therapeutic strategies from solely targeting amyloid proteins to multi-target interventions in AD.

1. Introduction

Alzheimer's disease (AD), the most common neurodegenerative dementia worldwide, has emerged as a formidable global health challenge, currently ranking as the fifth leading cause of mortality in individuals

aged ≥ 65 years and imposing profound socioeconomic burdens on healthcare systems across aging populations (Association, 2024). The clinical trajectory of AD typically commences with insidious short-term memory impairment that progressively undermines instrumental activities of daily living. Subsequent disease evolution manifests as a

Abbreviations: AD, Alzheimer's disease; BBB, Blood-brain barrier; A β , Amyloid- β ; DCE-MRI, Dynamic contrast-enhanced magnetic resonance imaging; DGE, Dynamic glucose-enhanced; PET, Positron emission tomography; TJ, tight junction; AQP4, Aquaporin-4; LRP1, Lipoprotein receptor-related protein 1; P-gp, P-glycoprotein; RAGE, Receptor for advanced glycation end products; GLUT1, Glucose transporter 1; APOE, Apolipoprotein E; CSF, Cerebrospinal fluid; ASL, Arterial spin labeling; CEST, Chemical exchange saturation transfer; FDG, Fluorodeoxyglucose.

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multifaceted cognitive syndrome encompassing temporal-spatial disorientation, executive dysfunction, linguistic deterioration, and neuropsychiatric disturbances, ultimately culminating in complete functional dependency during terminal stages (Scheltens et al., 2016). This insidious neurodegenerative process spans decades, evolving through three clinically distinct phases: (1) preclinical pathological accumulation, (2) prodromal mild cognitive impairment, and (3) dementia with irreversible neurological devastation (Scheltens et al., 2016). The 2011 revised diagnostic criteria revolutionized AD conceptualization by incorporating biomarker evidence, thereby establishing a biological definition that transcends purely syndromic characterization (McKhann et al., 2011). While AD neuropathology features well-characterized hallmarks including blood-brain barrier (BBB) disruption, neuroinflammatory activation, amyloid- β ($A\beta$) plaque deposition, tau neurofibrillary tangle formation, and synaptic/neuronal loss, the precise mechanistic interplay between these pathological processes and their relative contributions to cognitive decline remain contentious.

Since its inception three decades ago, the amyloid cascade hypothesis has dominated therapeutic development paradigms, positing extracellular $A\beta$ aggregates and intraneuronal hyperphosphorylated tau pathology as cardinal initiating events in AD pathogenesis (Braak and Del Tredici, 2015; Hardy and Higgins, 1992). According to this framework, these primary proteinopathies trigger secondary neurodegenerative cascades involving neurotoxic astrogliosis, microglial activation, synaptic stripping, axonal degeneration, and ultimately cortico-subcortical atrophy with white matter disintegration (Heneka et al., 2015; Jack et al., 2018; Shinohara et al., 2016; Spires-Jones and Hyman, 2014; Zlokovic, 2011). Despite generating crucial insights into $A\beta$ metabolism and facilitating neuropathological staging systems, the hypothesis's clinical translatability has been unexpectedly limited, with numerous anti-amyloid therapies demonstrating marginal clinical efficacy despite successful target engagement. This therapeutic impasse underscores critical knowledge gaps regarding AD's pathophysiological underpinnings, particularly the temporal-spatial relationships between molecular pathologies and their synergistic neurotoxic effects. Notably, while cerebral $A\beta$ dyshomeostasis represents an early pathogenic event, the fundamental mechanisms driving its accumulation remain elusive. Mounting evidence suggests impaired $A\beta$ clearance, rather than overproduction, disrupts cerebral metabolic equilibrium, favoring progressive accumulation of neurotoxic $A\beta$ species (Mawuenyega et al., 2010). Among multiple clearance pathways, BBB-mediated efflux via receptor-mediated transcytosis constitutes the principal elimination route, highlighting the imperative to elucidate BBB's multifaceted roles in AD pathogenesis.

The BBB, a sophisticated neurovascular interface comprising specialized endothelial cells, pericytes, and astrocytic end-feet, maintains cerebral homeostasis through selective molecular exchange between systemic circulation and neural parenchyma (Engelhardt, 2003). Beyond passive barrier functions, this dynamic structure actively regulates micronutrient transport, metabolic waste clearance, and immunological surveillance while maintaining strict ionic gradients essential for neuronal signaling (Daneman and Prat, 2015). Emerging research implicates BBB breakdown as both consequence and catalyst of AD progression, creating self-perpetuating neurodegenerative loops through impaired $A\beta$ clearance, inflammatory cell infiltration, and neurovascular uncoupling (Sweeney et al., 2018). However, the BBB's functional complexity (particularly its polarized transporter systems, heterocellular signaling networks, and region-specific permeability characteristics) presents formidable challenges for *in vitro* modeling. Revolutionary advances in multimodal neuroimaging now enable noninvasive *in vivo* interrogation of BBB integrity across the AD continuum, from asymptomatic genetic carriers to end-stage dementia patients (Montagne et al., 2016). These techniques allow systematic characterization of BBB alterations across the entire AD continuum, from presymptomatic carriers of genetic risk factors to patients with

advanced dementia. Modern imaging protocols combining high spatial resolution with longitudinal acquisition capabilities permit: (1) detection of subtle BBB leakage in preclinical stages, (2) spatial mapping of barrier dysfunction relative to classical AD pathologies, and (3) dynamic monitoring of neurovascular changes during therapeutic interventions. Particularly noteworthy are radiation-free MRI techniques that facilitate repeated assessments in vulnerable populations, allowing establishment of causal-temporal relationships between BBB deterioration and cognitive decline.

This review provides a critical synthesis of PET and MRI methodologies for BBB assessment in AD research (Fig. 1). We first delineate the BBB's neuroanatomical constituents and homeostatic functions, followed by systematic analysis of AD-induced barrier dysfunction mechanisms. Subsequently, we conduct a comprehensive evaluation of current imaging techniques and emphasized main findings, including dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI), water-exchange MRI techniques, dynamic glucose-enhanced (DGE) MRI, and positron emission tomography (PET) tracers targeting endothelial receptors. Finally, we propose a translational framework for optimizing BBB imaging biomarkers, emphasizing technical standardization needs, multimodal integration strategies, and artificial intelligence-powered analytical pipelines to advance early diagnosis and personalized therapeutic approaches.

2. BBB structure and function

The BBB represents a sophisticated multicellular vascular interface characterized by its selective permeability (Fig. 2), comprising three principal components: (1) cerebral microvascular endothelial cells forming a continuous tubular structure with unique barrier properties, (2) an intricate basement membrane matrix encapsulating the abluminal endothelial surface, and (3) perivascular support cells including pericytes, astrocyte end-feet projections, and neural networks that collectively maintain neurovascular homeostasis (Daneman and Prat, 2015; Erickson and Banks, 2013; Zenaro et al., 2017). Cerebral microvascular endothelial cells constitute the central element of the BBB, establishing a polarized cellular barrier that orchestrates precise molecular trafficking between systemic circulation and the central nervous system (Daneman, 2012; Zlokovic, 2008). Notably, these specialized endothelial cells exhibit three distinctive barrier-enhancing features: absence of fenestrations, abundant mitochondria supporting high energy demands, and complex tight junction (TJ) networks that effectively restrict paracellular diffusion (Daneman, 2012; Kadry et al., 2020). Mitochondrial dysfunction contributes to BBB impairment and leakage through multiple mechanisms (Kazemeini et al., 2024): (1) disrupting endothelial energy metabolic homeostasis (resulting in aberrant expression of tight junction proteins claudin-5 and occludin), (2) impairing ATP-dependent cytoskeletal remodeling and vesicular trafficking systems (thereby exacerbating the leakage of $A\beta$ -associated neurotoxic substances), and (3) inducing reactive oxygen species overproduction cascades (disrupting calcium homeostasis within the neurovascular unit and promoting endothelial apoptosis). Molecular characterization reveals that TJ assembly requires coordinated interactions between transmembrane proteins [claudins (Günzel and Yu, 2013), occludins (Furuse et al., 1993), and junctional adhesion molecules (Dejana et al., 2008)] and cytoplasmic scaffolding proteins such as zonula occludens-1. This dynamic junctional architecture, coupled with calcium-dependent adherens junctions, establishes a size-selective (<400 Da) and charge-dependent molecular gate (Dejana et al., 2008; Lochhead et al., 2020). Substance translocation across the BBB occurs through four principal mechanisms: (1) lipid-mediated free diffusion of small lipophilic molecules, (2) carrier-mediated transport of nutrients and ions, (3) receptor-mediated transcytosis for macromolecules, and 4) adsorptive-mediated transcytosis for cationic molecules (Deeken and Löscher, 2007; Wong et al., 2013). Embedded within the vascular wall, pericytes exhibit remarkable anatomical intimacy with endothelial cells, with 1:1 cellular ratio

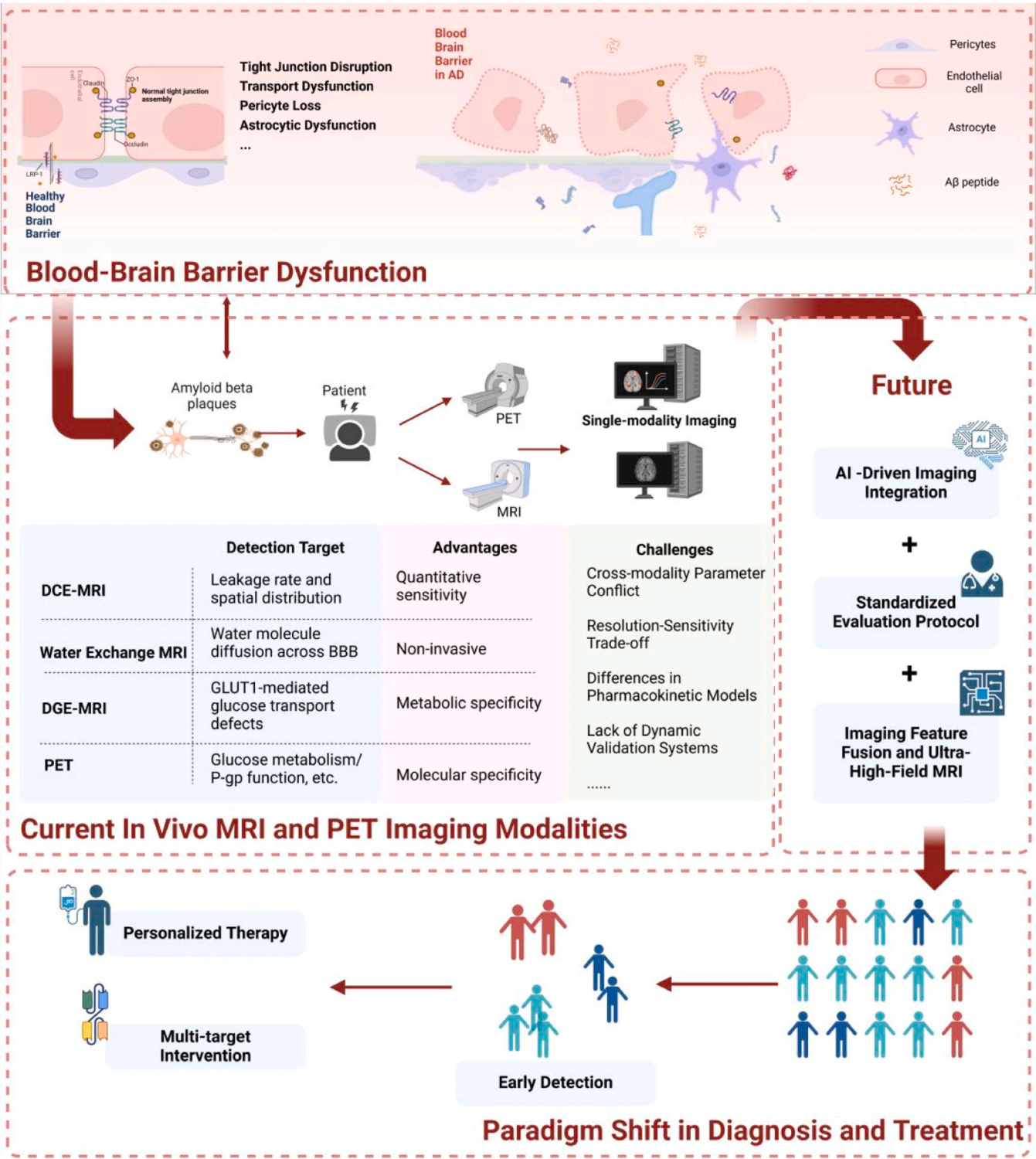


Fig. 1. Blood-Brain Barrier Dysfunction and Non-Invasive In Vivo Imaging Techniques. This figure presents the overall framework of the review content in this work. It emphasizes the non - invasive in vivo imaging techniques involved in the review that are used to detect blood - brain barrier dysfunction in Alzheimer’s disease. It also showcases the advantages and limitations of current multimodal imaging techniques when detecting blood-brain barrier dysfunction. Furthermore, future directions are proposed, including the need for technical standardization, multimodal integration strategies, and artificial intelligence - driven analysis processes. These directions aim to promote the development of early diagnosis and personalized treatment plans, with the ultimate goal of facilitating a transformation in the diagnosis and treatment paradigm. MRI= magnetic resonance imaging, DCE=Dynamic contrast-enhanced, DGE=Dynamic glucose-enhanced, PET=positron emission tomography. (Figure created using BioRender.com).

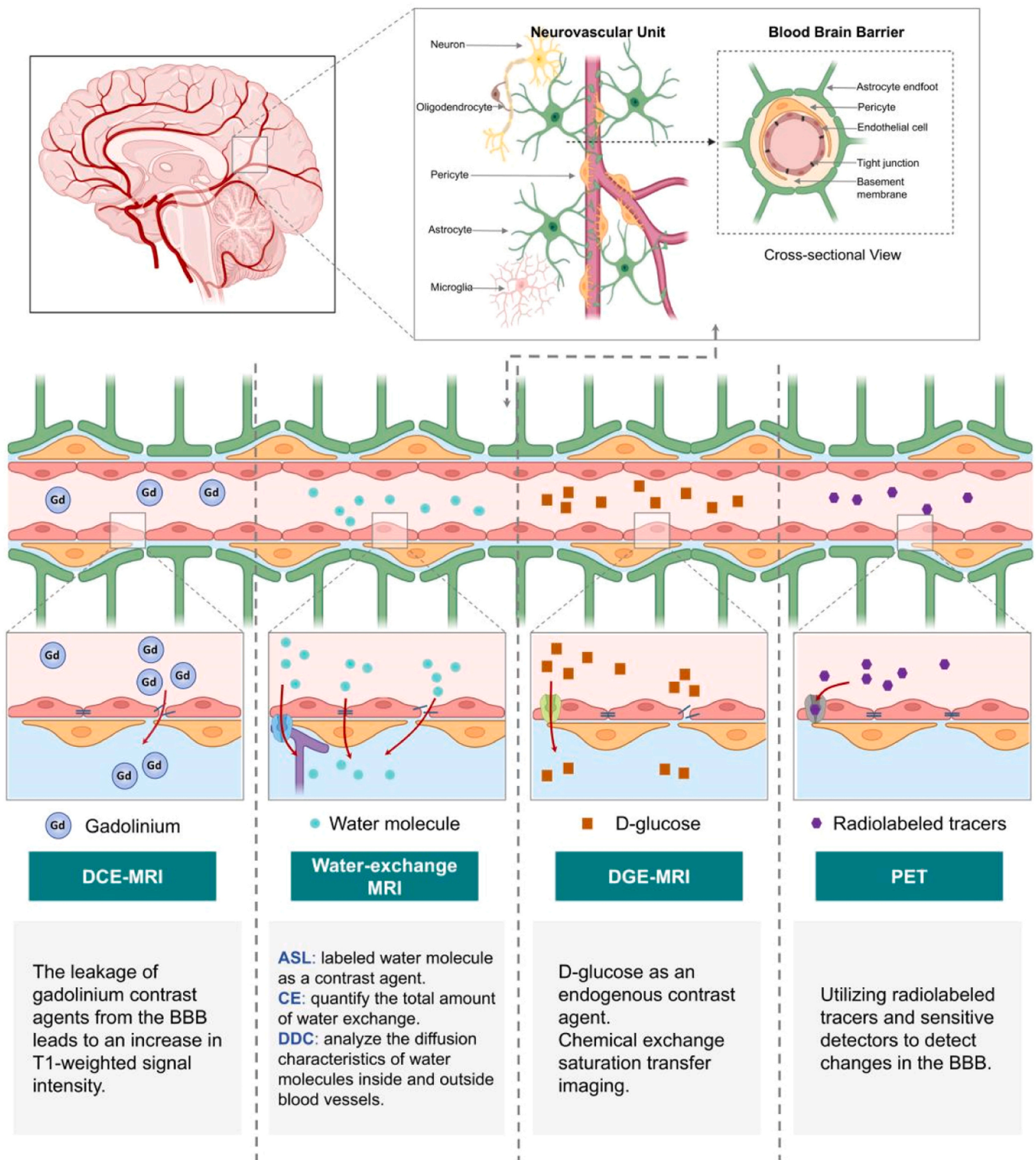


Fig. 2. Structural and functional characterization of the BBB and advanced neuroimaging modalities for detecting BBB pathophysiology. The neurovascular unit comprises tightly joined endothelial cells forming the primary barrier, supported by contractile pericytes embedded within the basement membrane, and ensheathed by astrocytic end-feet projections. In Alzheimer's disease pathogenesis, this intricate cellular architecture undergoes progressive disruption, manifesting as both increased paracellular permeability and dysregulation of specific transporter systems. Cutting-edge imaging techniques enable multidimensional assessment of BBB integrity: DCE-MRI provides quantitative assessment of BBB permeability through gadolinium-based tracer kinetics; Water exchange MRI offers non-invasive quantification of water molecule exchange rates across endothelial membranes; DGE imaging maps cerebral glucose uptake dynamics via chemical exchange saturation transfer mechanisms; Multitracer PET delineates specific transporter dysfunction using radiolabeled substrates for P-glycoprotein and other critical efflux pumps. BBB=blood-brain barrier, DCE-MRI=Dynamic contrast-enhanced MRI, DGE=Dynamic glucose-enhanced, PET=positron emission tomography, ASL=arterial spin labeling, CE=contrast-enhanced, DDC=double-diffusion encoding. (Figure created using BioRender.com).

observed in cerebral microvessels (Shepro and Morel, 1993). These multifunctional Pericytes execute four critical barrier functions (Armulik et al., 2010; Daneman et al., 2010; Knox et al., 2022; Winkler et al., 2011): (1) angiogenic regulation through vascular endothelial growth factor signaling modulation, (2) cerebrovascular tone control via contractile capacity, (3) extracellular matrix remodeling through metalloproteinase secretion, and (4) immune surveillance by regulating leukocyte diapedesis. Astrocytes extend polarized endfoot processes that enwrap > 99 % of the brain microvasculature, forming tripartite synapses with neurons and endothelial cells (Knox et al., 2022; Oberheim et al., 2009; Oksanen et al., 2019). This unique positioning enables astrocytes to perform dual regulatory roles: neurovascular coupling through Ca^{2+} -mediated vasoactive signaling (adjusting cerebral blood flow to neuronal activity) (Attwell et al., 2010; Sidoryk-Wegrzynowicz et al., 2010) and BBB maintenance via secretion of angioneurins like sonic hedgehog and glial-derived neurotrophic factor (Michinaga et al., 2021; Sweeney et al., 2019). Furthermore, astrocytic endfeet demonstrate polarized expression of homeostatic channels, including Kir4.1 potassium channels that buffer neuronal K^+ excess and aquaporin-4 (AQP4) water channels facilitating glymphatic clearance (Liu et al., 2018). The vascular basement membrane emerges as a dynamic interface rather than passive scaffolding, comprising two structurally distinct layers: the endothelial basement membrane (secreted by endothelial cells and pericytes) and the parenchymal basement membrane (produced by astrocytes) (Baeten and Akassoglou, 2011; Owens et al., 2008). This bilaminar structure not only provides mechanical support but also serves as a signaling platform through laminin-integrin interactions, a molecular sieve regulating leukocyte trafficking, and a rheostat for TJ protein localization through matrix metalloproteinase activation (Baeten and Akassoglou, 2011; Savettieri et al., 2000; Tilling et al., 1998).

The central nervous system employs multiple clearance systems for A β and tau protein homeostasis: intracellular degradation (ubiquitin-proteasome system, autophagy-lysosome pathway), extracellular proteolysis (neprilysin, insulin-degrading enzyme), and efflux pathways (BBB transport, glymphatic drainage, meningeal lymphatic clearance) (Xin et al., 2018). Notably, A β clearance exhibits pathway compartmentalization: soluble A β predominantly undergoes receptor-mediated BBB transcytosis, while aggregated forms rely more on perivascular drainage along arterial basement membranes (Xin et al., 2018). The BBB emerges as the principal regulator of A β flux, employing polarized transporter systems on endothelial cells: abluminal lipoprotein receptor-related protein 1 (LRP1) mediates brain-to-blood efflux, luminal P-glycoprotein (P-gp) completes transendothelial transport, while receptor for advanced glycation end products (RAGE) operates bidirectionally with net influx predominance (Cirrito, 2005; Sweeney et al., 2018; Zhao et al., 2015). In AD pathogenesis, this delicate transport equilibrium becomes disrupted through (1) RAGE upregulation enhancing A β influx (Yamazaki and Kanekiyo, 2017), (2) LRP1 downregulation impairing efflux capacity, and (3) P-gp functional decline reducing luminal export (Cirrito, 2005; Sweeney et al., 2018; Zhao et al., 2015). This transporter triad dysfunction creates a "perfect storm" for A β accumulation with reduction in net A β clearance efficiency during AD progression (Zhao et al., 2015).

3. BBB breakdown in AD

The precise mechanistic contribution of BBB dysfunction to AD pathogenesis remains incompletely characterized, though emerging evidence positions cerebrovascular impairment as both a contributor to and consequence of neurodegenerative processes. A β pathology and BBB deterioration appear to engage in reciprocal potentiation, establishing a self-reinforcing cycle that accelerates disease progression. At the molecular level, A β oligomers directly compromise endothelial barrier function through structural disintegration of TJ complexes. Convergent findings from postmortem human studies and transgenic models reveal

substantial reductions in key TJ components including occludin, claudin-5, and zonula occludens-1, with genetic ablation studies confirming the functional necessity of claudin-5 and occludin for maintaining junctional integrity (Carrano et al., 2011; Carrano et al., 2012; Kook et al., 2012; Wan et al., 2015; Yamazaki et al., 2019). Beyond structural components, BBB dysfunction manifests through impaired transport mechanisms. The glucose transporter 1 (GLUT1), a critical mediator of cerebral metabolic homeostasis, demonstrates marked downregulation in both AD patients and murine models (Hooijmans et al., 2007; Merlini et al., 2011). This deficiency initiates a pathogenic cascade characterized by impaired glucose utilization, TJ protein depletion, microvascular degeneration, and exacerbated A β accumulation with concomitant cognitive decline (Kyrtata et al., 2021; Winkler et al., 2015). Notably, GLUT1 deficiency exerts secondary effects on A β clearance through transcriptional suppression of LRP1, establishing a dual mechanism linking metabolic insufficiency to amyloid pathology (Winkler et al., 2015). Cellular contributors to the neurovascular unit exhibit distinct vulnerability patterns in AD. Pericytes, the sentinel regulators of microvascular stability, demonstrate reductions in cerebrovascular coverage in AD brains (Sengillo et al., 2013; Sweeney et al., 2016). Experimental depletion models recapitulate key AD features including amplified A β deposition, accelerated cerebral amyloid angiopathy progression, and secondary neurodegenerative changes (Bell et al., 2010). Astrocytic dysfunction, particularly in water homeostasis regulation, manifests through impaired glymphatic clearance and disrupted AQP4 polarization (Iliff et al., 2012; Lan et al., 2015; Valenza et al., 2020). Clinical-pathological correlations reveal striking reductions in perivascular AQP4 localization that strongly associate with both A β burden and Braak staging (Zeppenfeld et al., 2017), while mouse models demonstrate A β -induced astrocytic endfoot retraction and AQP4 mislocalization (Merlini et al., 2011; Yang et al., 2011). Structural remodeling of the vascular basement membrane, characterized by collagen IV, perlecan, and fibronectin accumulation (Farkas and Luiten, 2001), likely impairs perivascular clearance pathways critical for A β removal (Arai et al., 2012; Hawkes et al., 2013; Morris et al., 2016).

AD is classified into familial and sporadic subtypes (Long and Holtzman, 2019). Sporadic cases predominantly manifest as late-onset AD, primarily affecting individuals over 65 years of age, with the Apolipoprotein E (APOE) $\epsilon 4$ allele identified as the strongest genetic risk factor. Familial AD (FAD) exhibits autosomal dominant inheritance patterns as early-onset AD, accounting for 1 %-5 % of total cases, characterized by premature cognitive decline with symptom onset between 30 and 65 years. This subtype is primarily driven by pathogenic mutations in three genes: presenilin 1 (PSEN1), presenilin 2 (PSEN2), and amyloid precursor protein. A substantial number of genes associated with AD risk are functionally implicated in the cerebral vascular and perivascular systems, highlighting the critical role of BBB integrity in the disease's pathological progression. Genome-wide association studies have revealed that 30 out of the top 45 AD risk-associated genes are expressed in the cerebral vasculature (Yang et al., 2022). The extent of BBB impairment is exacerbated in APOE $\epsilon 4$ carriers. The ApoE $\epsilon 4$ allele represents the strongest genetic risk factor for sporadic late-onset AD (Halliday et al., 2016), with ApoE participating in amyloid plaque formation and influencing A β aggregation depending on its lipidation state (Dal Magro et al., 2019). The co-localization of ApoE with A β in pericytes and perivascular astrocytes suggests its potential involvement in A β transcytosis across the BBB (Blanchard et al., 2020; Utter et al., 2008). Histopathological examinations of APOE $\epsilon 4$ carriers demonstrate accelerated pericyte and BBB damage accompanied by increased accumulation of pro-inflammatory cytokines (Miners et al., 2018; Zipser et al., 2007), which correlates with MRI-observed hippocampal BBB permeability changes independent of tau and A β pathology (Montagne et al., 2020). The phosphatidylinositol binding clathrin assembly protein gene protects BBB integrity against A β toxicity through restoration of clathrin-mediated endocytosis (Juul Rasmussen et al., 2019) and promotion of amyloid precursor protein autophagic degradation (Tian et al.,

2013). Other genetic factors, including bridging integrator 1 and CD2-associated protein, influence AD progression by modulating A β transcytosis across the BBB (Lambert et al., 2013). The Ras and Rab interactor 3 gene, involved in vesicular trafficking, has been implicated in BBB dysfunction (Kajiho et al., 2003). Triggering receptor expressed on myeloid cells 2, an AD susceptibility gene encoding triggering receptor expressed on myeloid cells 2, primarily expressed in brain microglia, is associated with neuroinflammatory responses and BBB dysfunction (Jonsson et al., 2013; Winfree et al., 2022). Soluble triggering receptor expressed on myeloid cells 2 may compromise BBB integrity through interactions with pro-inflammatory proteins (Rauchmann et al., 2020). BBB integrity and function are further maintained by additional genetic regulators. The OCLN gene encoding occludin and JAM-C gene encoding junctional adhesion molecule-C are crucial for BBB integrity (Akawi et al., 2013; Wyss et al., 2012). Loss-of-function mutations in major facilitator superfamily domain containing 2a, which mediates ω -3 fatty acid transport, and NOTCH3, essential for smooth muscle cell differentiation, significantly impact BBB structural and functional stability (Ben-Zvi et al., 2014; Chabriot et al., 2009). Formin like 2 gene, another AD genetic risk factor, shows disease-state-dependent expression in astrocytes, regulating glial-vascular interactions and A β peptide modulation, with elevated levels observed in AD patients (Lee et al., 2022). Deletion of uncoordinated-5 homolog B gene has been shown to reduce Wnt/ β -catenin signaling activity in the BBB (Boyé et al., 2022). In the pathological mechanisms of FAD, mutations in amyloid precursor protein, PSEN1, and PSEN2 genes promote abnormal aggregation of A β (Krishnamurthy et al., 2025), exacerbating neurovascular inflammatory responses and disrupting transport systems (e.g., LRP1 and P-gp), ultimately leading to BBB dysfunction and accelerated degradation (Cortes-Canteli and Iadecola, 2020). Additionally, several studies have demonstrated a potential A β -independent association between classical FAD-associated genes and BBB dysfunction. Early vascular remodeling observed in mouse models carrying mutant PSEN1 or PSEN2 may involve A β oligomer-triggered compensatory responses or neurotrophic signaling abnormalities independent of A β that directly promote vascular restructuring (Gama Sosa et al., 2010). In FAD patient-derived induced pluripotent stem cell-differentiated brain microvascular endothelial cells, PSEN1 mutations demonstrate more pronounced BBB integrity disruption compared to PSEN2 mutations, manifesting as tight junction structural abnormalities, reduced drug efflux pump activity, and glucose metabolism dysregulation (Raut et al., 2021). Furthermore, environmental stressors such as hypoxia and glucose deprivation specifically upregulate PSEN2 expression levels, suggesting this gene may play a distinct regulatory role in vascular stress responses (Mohuczy et al., 2002).

The expression levels of TJ proteins are closely associated with AD pathological features. A β 42-mediated TJ disruption occurs through RAGE receptor interaction and calcium-calmodulin signaling pathway activation, which can be mitigated by neutralizing antibodies against RAGE and calcineurin, along with matrix metalloproteinase inhibitors (Kook et al., 2012). BBB disruption is also linked to Wnt/ β -catenin signaling dysfunction, which is crucial for maintaining BBB integrity through regulation of TJ protein expression and transporter activity (Liu et al., 2014), affecting both paracellular and transcellular permeability (Hussain et al., 2022). A β interactions with astrocytes and microglia activate the NLR family pyrin domain containing 3 inflammasome, triggering inflammatory mediator release and caspase-1 cascade activation, ultimately compromising BBB integrity (Liang et al., 2022). NF- κ B, activated by A β plaques and other factors, promotes pro-inflammatory cytokine and reactive oxygen species production, exacerbating neuroinflammation and oxidative stress (Al-Ghraiyyah et al., 2022), while also activating matrix metalloproteinases (MMPs) and disrupting redox-sensitive transcription factors to impair BBB function (Lehner et al., 2011). In pericytes, activation of the pro-inflammatory Cyclophilin A-MMP9 pathway contributes to BBB integrity loss, with ApoE4 variants enhancing Cyclophilin A-MMP9

activity (Bell et al., 2012). NF- κ B-mediated transcription upregulates cerebrovascular MMP9 expression, leading to BBB dysfunction (Candelario-Jalil et al., 2011). A β -mediated pericyte signaling induces vasoconstriction and cerebral blood flow reduction (Nortley et al., 2019), while A β deposition causes pericyte degeneration (Ding et al., 2020), both contributing to BBB impairment. ApoE4 disrupts BBB through calcineurin-nuclear factor of activated T-cells signaling pathway interference (Blanchard et al., 2020), and reduced platelet-derived growth factor receptor β signaling with decreased pericyte proliferation further compromises BBB in AD. The 5'-AMP-activated protein kinase signaling pathway is essential for maintaining BBB integrity, particularly under neuroinflammatory conditions, with reduced activation leading to barrier impairment, while 5'-AMP-activated protein kinase activators like metformin can ameliorate such damage (Wang et al., 2017). Additionally, increased intracellular cyclic AMP levels enhance BBB function through direct upregulation of TJ protein-5 expression or its phosphorylation via the cAMP/PKA pathway (Takata et al., 2021).

In AD, females exhibit a heightened susceptibility, manifesting as disproportionate burdens in disease prevalence, progression, and clinical phenotypes (Ferretti et al., 2018; Mielke et al., 2022). Both clinical and preclinical evidence indicate that BBB dysfunction exhibits significant sex-related disparities. Brain microvascular endothelial cells differentiated from female induced pluripotent stem cells demonstrate higher transendothelial electrical resistance (indicative of lower BBB permeability) and elevated GLUT1 expression compared to male-derived cells (Lippmann et al., 2012). Clinical investigations further reveal that female subjects exhibit significantly lower levels of cerebrospinal fluid biomarkers reflecting BBB disruption and vascular damage, including cerebrospinal fluid (CSF)/serum albumin ratio, VCAM1, and E-selectin (Castellazzi et al., 2020; Li et al., 2017; Parra-do-Fernández et al., 2018). However, females display more pronounced age-related declines in BBB metabolic activity (Zhao et al., 2016) and structural integrity (Moon et al., 2021). This heightened female vulnerability to BBB compromise may partially stem from hormonal influences, particularly the loss of estrogen's neurovascular protective effects during menopause and postmenopausal periods. Notably, ABCA1 gene mutations associated with dysfunction of ATP-binding cassette transporters at the BBB exclusively elevate AD risk in females (Sundar et al., 2007). These sex-specific manifestations of BBB dysfunction underscore the imperative to elucidate their pathophysiological mechanisms for comprehensive understanding of BBB contributions to AD pathogenesis. The potential mechanisms underlying these sex-related disparities in BBB dysfunction may involve multiple aspects, including oxidative stress, endothelial functional regulation, neuroinflammation, and immune responses (Cui et al., 2023; Weber and Clyne, 2021).

Estrogen preserves BBB integrity and endothelial function through multifaceted mechanisms. Preclinical studies demonstrate its protective effects via reduction of mitochondrial reactive oxygen species generation in cerebral vasculature, a process associated with maintenance of tight junction protein expression (Razmara et al., 2007). Its stabilizing effects on the BBB involve dual regulation of extracellular matrix and cytoskeletal dynamics: modulating matrix metalloproteinase activity to prevent structural degradation (Rempe et al., 2016), while inhibiting RhoA/ROCK-2 signaling to counteract pathological cytoskeletal remodeling (Feng et al., 2018). Through membrane-initiated signaling, estrogen enhances tight junction stability via annexin A1 phosphorylation-mediated activation of formyl peptide receptor 2 (Maggioli et al., 2016). At the endothelial regulatory level, estrogen engages both classical receptors and G protein-coupled receptors, activating both genomic and non-genomic pathways to upregulate endothelial nitric oxide synthase expression and activity, thereby enhancing nitric oxide-mediated vascular regulation (Iorga et al., 2017). This nitric oxide-dependent mechanism may exhibit more pronounced clinical significance in postmenopausal women, as estrogen decline reduces nitric oxide synthase expression, accelerating BBB dysfunction (Mannacio

et al., 2012). Sex-dimorphic BBB dysfunction may also correlate with heterogeneous glial activation patterns. Clinical evidence reveals significantly higher CSF levels of the astrocytic/microglial activation marker YKL-40 in female AD patients compared to males, a disparity persisting across both healthy aging and AD pathological stages (Salvado et al., 2021). Animal models further demonstrate that female APP/PS1 transgenic mice exhibit amplified microglial activation phenotypes with sustained elevation of proinflammatory cytokines and complement pathway activity during aging (Biechele et al., 2020; Guillot-Sestier et al., 2021). In vitro studies at the cellular level reveal that female rat hippocampal astrocytes exhibit stronger proinflammatory polarization in response to A β 1–40 stimulation, whereas male counterparts demonstrate proliferative advantages and apoptosis-resistant properties (Lennol et al., 2021). Collectively, these findings suggest that sex-dependent glial responses to pathological stimuli may constitute a critical molecular basis for the observed sexual dimorphism in AD risk and BBB dysfunction.

While the CSF albumin quotient remains a conventional metric for evaluating BBB integrity, its clinical utility as a predictive biomarker is constrained by two fundamental limitations (Blennow et al., 1990; Erickson and Banks, 2019; Galea, 2021). First, its dependence on passive transendothelial transport of large plasma proteins renders it insensitive to subtle but clinically significant barrier alterations. Second, the non-specific nature of albumin leakage fails to distinguish between distinct molecular pathways underlying various neuropathological processes. This diagnostic ambiguity has spurred investigations into more mechanistically informative biomarkers, revealing a spectrum of molecular signatures associated with specific components of neurovascular unit dysfunction. Proteolytic disruption of the basal lamina manifests through elevated active matrix metalloproteinase-9 and its upstream regulator, the proinflammatory cytokine cyclophilin A (Bell et al., 2012; Halliday et al., 2016; Ryu and McLarnon, 2009). Concurrently, extravasation of hemostatic proteins like fibrinogen and plasminogen provides direct evidence of vascular leakage (Bell et al., 2012; Halliday et al., 2016; Ryu and McLarnon, 2009). At the endothelial interface, upregulated expression of intercellular adhesion molecule-1 and vascular cell adhesion molecule-1 reflects inflammatory activation and leukocyte trafficking potential (Nielsen et al., 2007). Pericyte degeneration, a critical event in microvascular instability, is marked by increased soluble platelet-derived growth factor receptor- β in biofluids (Cicognola et al., 2023; Montagne et al., 2015). Astrocytic responses to barrier compromise are detectable through elevated CSF levels of S100 calcium-binding protein B, observed in subsets of AD patients (Mori et al., 2010; Noijien et al., 1997; Peskind et al., 2001). Emerging blood-based assays attempt to profile neurovascular unit integrity through detection of S100B, ubiquitin carboxyl-terminal hydrolase L1, neuron-specific enolase, adhesion molecules, and glial fibrillary acidic protein (Mir and Chalak, 2014; Zuliani et al., 2008). Despite these advances in molecular characterization, current biomarker strategies face three principal translational challenges. First, CSF analysis requires invasive lumbar puncture procedures that limit patient compliance and preclude repeated sampling for longitudinal assessment. Second, peripheral blood biomarkers predominantly reflect systemic vascular status rather than specific cerebrovascular pathology due to substantial peripheral contributions and differential blood-CSF partitioning. Third, both approaches provide temporally static and anatomically nonspecific readouts, lacking spatial resolution to localize BBB defects within specific neuroanatomical regions and temporal resolution to track dynamic permeability changes. Crucially, existing biomarkers predominantly detect downstream consequences of barrier failure rather than directly interrogating structural and functional properties of the neurovascular unit. To address these limitations, next-generation diagnostic platforms must integrate two critical capabilities: (1) non-invasive imaging modalities capable of mapping BBB integrity with high spatiotemporal resolution across disease-relevant brain regions, and (2) multi-parametric analytical frameworks that correlate localized barrier

pathology with region-specific neurodegenerative patterns. Such technological integration could elucidate the spatiotemporal trajectory of BBB disruption in AD progression and its mechanistic interplay with A β deposition and tau propagation.

4. Current in vivo MRI and PET techniques in BBB pathophysiology

The integrity of the BBB represents a critical frontier in understanding AD pathogenesis, with advanced neuroimaging emerging as a cornerstone for noninvasive assessment of its structural and functional alterations. Advances in neuroimaging have revolutionized the noninvasive assessment of BBB integrity, offering critical insights into its role in AD pathophysiology. MRI and PET stand at the forefront of these innovations, employing distinct mechanisms to probe structural, functional, and molecular BBB alterations. DCE-MRI has established itself as the clinical standard for detecting subtle BBB leakage through gadolinium-based tracer kinetics, enabling quantitative characterization of permeability via parameters like K_{trans}. Parallel innovations in water-exchange MRI techniques [arterial spin labeling (ASL), contrast-enhanced protocols, and double diffusion encoding] provide gadolinium-free alternatives to evaluate trans-BBB water dynamics, reflecting both passive diffusion and active transport mechanisms. Meanwhile, DGE-MRI capitalizes on endogenous glucose as a chemical exchange saturation transfer (CEST)-based contrast agent to map BBB glucose transport deficiencies linked to GLUT1 dysfunction in AD models. On the molecular imaging front, PET methodologies employ radiotracers such as [18 F]-fluorodeoxyglucose (FDG) and [11 C]-verapamil to quantify glucose metabolic derangements and P-gp efflux impairments, respectively, while emerging tracers target aquaporins, RAGE receptors, and neuroinflammatory markers. The integration of multiparametric MRI with multitracers PET approaches promises to disentangle the spatiotemporal complexity of BBB disruption, offering a multidimensional framework to decode its role in AD progression, therapeutic response, and personalized biomarker development. Together, these modalities illuminate diverse aspects of BBB pathology (Fig. 2)—from subtle leakage and impaired clearance to transporter dysfunction—while addressing complementary challenges in spatial resolution, sensitivity, and molecular specificity. This section synthesizes the principles, applications, and evolving contributions of MRI and PET in mapping BBB disruptions across the AD continuum, highlighting their potential to unravel mechanistic links between BBB dysfunction and AD.

4.1. Magnetic resonance imaging

MRI has emerged as a pivotal neuroimaging modality for noninvasive assessment of BBB alterations in AD. This technique capitalizes on the properties of hydrogen nuclei in biological tissues, which undergo resonance excitation when exposed to radiofrequency pulses within a strong static magnetic field (typically 1.5–7 Tesla). The resultant nuclear magnetic resonance signals, characterized by distinct T1 and T2 relaxation properties, are spatially encoded through precisely controlled gradient fields to generate high-resolution anatomical and functional images (Arani et al., 2024). The non-ionizing nature of MRI coupled with its superior soft tissue contrast makes it particularly advantageous for serial monitoring of BBB integrity across different disease stages. The technique offers distinct advantages for BBB evaluation through three principal mechanisms: (1) High spatial resolution (sub-millimeter scale) enabling precise localization of subtle BBB defects; (2) Multimodal acquisition protocols sensitive to diverse physiological parameters including vascular permeability, molecular transport, and hemodynamic changes; (3) Elimination of radiation exposure risks associated with PET or computed tomography. Advanced MRI techniques have been specifically optimized for BBB characterization: Water exchange MRI quantifies transendothelial water permeability through

intravascular/extracellular compartment analysis; DCE-MRI tracks gadolinium-based tracer extravasation kinetics; DGE-MRI monitors cerebral glucose transport dynamics via CEST mechanisms. The integration of these complementary techniques in multiparametric MRI protocols permits comprehensive characterization of BBB pathophysiology, ranging from initial endothelial dysfunction to advanced structural breakdown in AD progression.

4.1.1. DCE-MRI: Quantifying Permeability via Gadolinium Kinetics

DCE-MRI has emerged as a cornerstone imaging modality for non-invasive evaluation of BBB integrity compromise, as schematically illustrated in Fig. 3. This quantitative technique employs sequential T1-weighted acquisitions following intravenous bolus administration of gadolinium-based contrast agents, capturing temporal evolution of contrast agent distribution within cerebral tissues. Longitudinal signal enhancement patterns, arising from paramagnetic gadolinium-induced T1 shortening effects, are systematically analyzed through advanced pharmacokinetic modeling approaches to derive quantitative permeability parameters (Uchida et al., 2023; Varatharaj et al., 2018). The fundamental principle underlying this methodology relies on the restricted biodistribution of gadolinium chelates in neurovascular units with intact BBB function. Pathological alterations in BBB permeability permit tracer extravasation into the perivascular extracellular space, with subsequent accumulation proportional to the degree of barrier

disruption. Spatial-temporal analysis of contrast agent kinetics enables precise topographic mapping and severity stratification of BBB permeability alterations. Current research applications have extensively utilized this paradigm to investigate neurovascular unit dysfunction across the AD continuum. Methodological implementations and key findings from contemporary DCE-MRI studies in AD populations are comprehensively summarized in Table 1.

Building upon these foundational observations, earlier studies used semi-quantitative analyses, revealing increased hippocampal enhancement in mild cognitive impairment (Wang et al., 2006a) and distinct patterns in AD versus non-demented elderly (Starr et al., 2009). To address methodological limitations, subsequent research employed more quantitative pharmacokinetic modeling to derive the volume transfer constant (Ktrans) representing permeability (Sourbron and Buckley, 2013). Critically, accurately quantifying leakage requires correcting for contrast delivery via blood flow. Given the minimal reflux characteristics observed in AD pathophysiology, the Patlak model demonstrates superior suitability over Tofts or extended Tofts models when assessing slight BBB leakage in AD populations (Barnes et al., 2016; Cramer and Larsson, 2014; Patlak and Blasberg, 1985; Tofts et al., 1999). This model's theoretical framework rests on six key assumptions: (1) Two compartments (the extravascular extracellular compartment and the vascular compartment) are separated by the BBB; (2) The diffusion of contrast agent is mainly limited by permeability; (3) The

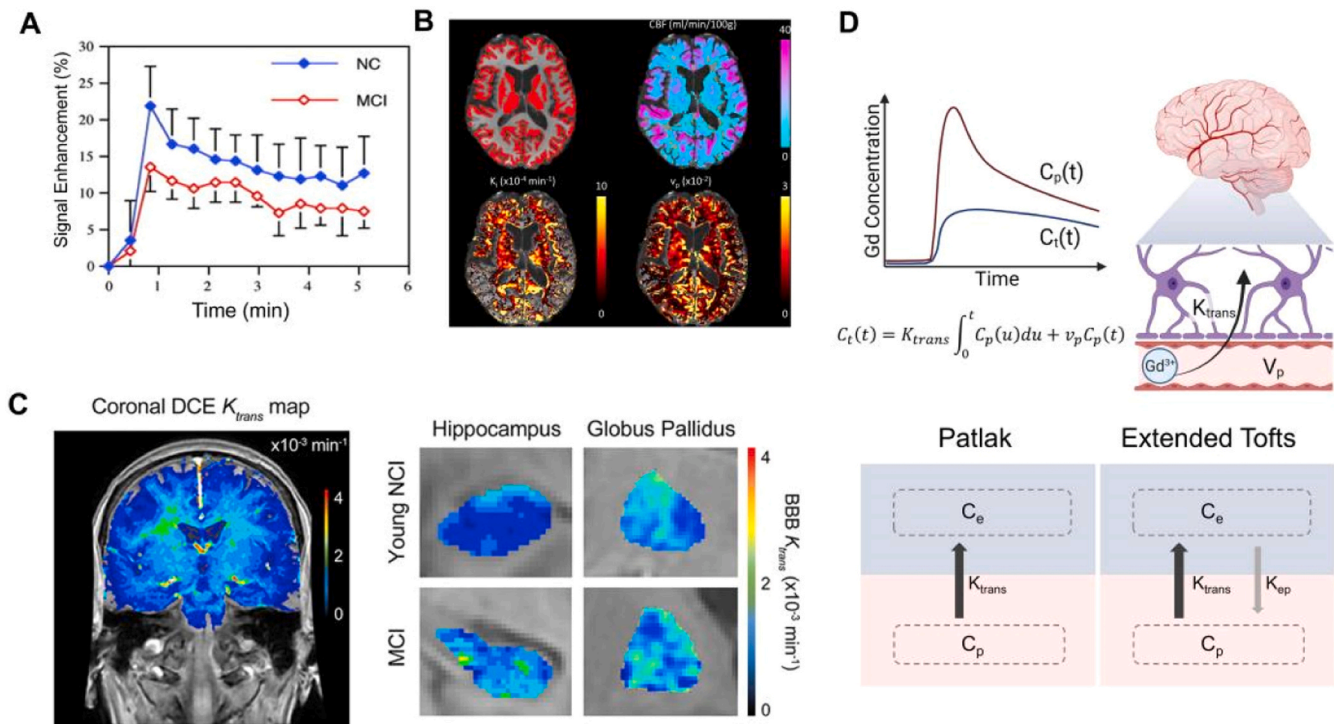


Fig. 3. A. Initial semi-quantitative analysis of gadolinium-based contrast agent dynamics through time-signal curve evaluation revealed distinct temporal patterns: MCI patients demonstrated a reduction in early-phase enhancement accompanied by prolonged signal attenuation within hippocampal regions compared to cognitively normal controls, suggesting altered contrast agent pharmacokinetics (Obtained from Wang et al. (Wang et al., 2006a)). B. Quantitative parametric analysis employing DCE-MRI methodology provides enhanced physiological specificity through three principal biomarkers: 1) BBB leakage rate (K_i , mL/100 g/min), 2) Plasma volume fraction (V_p , %), and 3) Cerebral blood flow (CBF, mL/100 g/min). These metrics enable precise quantification of neurovascular unit dysfunction across different dementia stages. (Obtained from van de Haar et al. (van de Haar et al., 2016b)). C. Parametric mapping visualization: Left panel displays whole-brain K_{trans} distribution in cognitively preserved elderly, demonstrating physiological BBB integrity. Right panel contrasts regional K_{trans} values between young NCI controls (upper row) and age-matched MCI patients (lower row), revealing higher permeability in hippocampal and pallidal regions in MCI subjects (Obtained from Montagne et al. (Montagne et al., 2019)). D. Pharmacokinetic modeling framework: DCE-MRI quantifies BBB integrity through sequential measurement of contrast agent concentration differentials between cerebral tissue (C_t) and plasma (C_p). The Patlak model applies to mild BBB disruption where tracer reflux is negligible (C_p : $C_e > 10:1$). For advanced BBB compromise, the extended Tofts model incorporates bidirectional exchange through the rate constant K_{ep} ($K_{ep} = K_{trans}/V_e$), where V_e represents extravascular extracellular space volume (Obtained from Chagnot et al. (Chagnot et al., 2021)). MCI = mild cognitive impairment, NCI = non-cognitive impairment, AD = Alzheimer's disease, K_{trans} = volume transfer constant, V_p = plasma volume, CBF = cerebral blood flow, K_i = blood-brain barrier leakage rate, DCE-MRI = dynamic contrast-enhanced magnetic resonance imaging, BBB = blood-brain barrier, C_t = tissue contrast concentration, C_p = plasma contrast concentration, K_{ep} = efflux rate constant, V_e = extravascular extracellular volume fraction.

Table 1

Overview of studies using DCE-MRI to assess blood-brain barrier permeability in AD continuum.

Study: author (year)	subject sample size	Scanner model and coil configuration	Imaging parameters	Main BBB findings
Wang et al. (2006a)	11 patients with MCI, 11 healthy elderly controls	Scanner model: Philips 1.5-Tesla Eclipse scanner Coil configuration: Not specified.	DCE-MRI scan parameters: T1-weighted 3D gradient echo sequence with 16 acquisitions, TR 10 msec, TE 3 msec, flip angle 20°, FOV 22 cm, matrix size 256 × 256, slice thickness 8 mm, temporal resolution 25.5 s. Contrast agent usage: Omniscan at a dose of 1 cm3/10 lb of body weight. Pharmacokinetic model: Not specified.	No specific values provided.
Starr et al. (2009)	15 AD patients, 15 healthy controls	Scanner model: GE Signa LX 1.5 T scanner, Coil configuration: quadrature birdcage head coil	DCE-MRI scan parameters: 3D FSPGR sequence, TR: 29 ms, TE: 3 ms, flip angle: 24°, acquisition time: 3 min 46 sec. Contrast agent administration: 20 mL Magnevist (gadopentetate dimeglumine) IV at 5 mL/s. Pharmacokinetic model: Not specified.	No specific values provided.
Montagne et al. (2015)	6 young NCI, 18 older NCI, 21 MCI	Scanner model: GE 3 T HDXT MR scanner Coil configuration: Not specified.	DCE-MRI scan parameters: 3D SPGR sequence, TR: 16 min, Temporal resolution: 15.4 sec per image, Voxel size: 0.625 × 0.625 × 5 mm. Contrast agent administration: MultiHance (gadobenate dimeglumine), 0.05 mmol/kg IV. Pharmacokinetic model: Modified Patlak model.	Patients BBB Ktrans($\times 10^{-3} \text{ min}^{-1}$): MCI: Cortex Superior Frontal Gyrus Cortex Superior Frontal Gyrus= 1.49 ± 0.31, Cortex Inferior Temporal Gyrus= 1.27 ± 0.25, Thalamus= 0.89 ± 0.24, Caudate Nucleus= 1.76 ± 0.31, Striatum= 1.85 ± 0.18, White Matter= 2.30 ± 0.36, Corpus Callosum= 2.19 ± 0.33, Internal Capsule= 2.02 ± 0.18.
van de Haar et al. (2016b)	7 AD, 7 MCI, 16 NCI	Scanner model: Philips Achieva 3.0 T scanner Coil configuration: 32-channel receiver head coil	DCE-MRI fast scan parameters: 3D saturation recovery gradient echo, TR/TE 5.2/2.5 ms, 29 volumes, interval 3.2 s DCE-MRI slow scan parameters: Saturation recovery gradient echo, TR/TE 5.6/2.5 ms, 45 volumes, interval 31.8 s Contrast agent: 0.1 mmol/kg gadobutrol IV Pharmacokinetic model: Patlak model	Patients BBB Ktrans($\times 10^{-3} \text{ min}^{-1}$): Whole gray matter= 2.7 ± 0.14
van de Haar et al. (2016a)	16 patients with early AD (7 AD, 9 MCI) 17 healthy control subjects	Scanner: 3 T Philips Achieva MRI Coil configuration: Dedicated head coil	DCE-MRI Protocol: Dual-temporal resolution Fast sequence: saturation recovery GRE, TR/TE 5.2/2.5 ms, temporal resolution 3.2 sec. Slow sequence: saturation recovery GRE, TR/TE 5.6/2.5 ms, temporal resolution 30.5 sec. 25 min total scan time. Contrast Agent: 0.1 mmol/kg gadobutrol injected at 3 mL/sec. Pharmacokinetic model: Patlak model.	Patients BBB Ktrans($\times 10^{-3} \text{ min}^{-1}$): Gray Matter= 0.89 ± 1.12 White Matter= 0.66 ± 0.44 Normal White Matter= 0.65 ± 0.43 Deep Gray Matter= 1.25 ± 1.15 Cortex= 0.84 ± 1.14 White Matter Hyperintensities= 1.06 ± 1.11
van de Haar et al. (2017)	16 early AD, 17 healthy controls	Scanner: 3 T Philips Achieva MRI Coil configuration: Dedicated head coil	DCE-MRI Protocol: Dual time resolution Fast: 3D gradient echo, TR/TE 5.2/2.5 ms, 3.2 s temporal resolution. Slow: 3D gradient echo, TR/TE 5.6/2.5 ms, 30.5 s temporal resolution. 25 min total scan time. Contrast Agent: Gadobutrol, 0.1 mmol/kg Pharmacokinetic Model: Patlak model.	Patients BBB Ktrans($\times 10^{-3} \text{ min}^{-1}$): White matter = 0.75 Gray matter = 1.04
Montagne et al. (2019)	40 NCI, 12 MCI	Scanner model: GE 3 T HDXT MR scanner Coil configuration: 8-channel head coil	DCE-MRI scan parameters: 3D SPGR sequence, TR: 16 min, Temporal resolution: 15.4 sec per image, Voxel size: 0.625 × 0.625 × 5 mm. Contrast agent administration: 0.05 mmol/kg cyclic gadoterate or linear gadobenate Pharmacokinetic model: Modified Patlak model.	No specific values provided.
Nation et al. (2019)	44 NCI, 23 MCI	Scanner model: 3 T GE HDXT MRI Coil configuration: 8-channel head coil	DCE-MRI scanning parameters: T1-weighted 3D SPGR pulse sequence: FA 15°, TR/TE 8.29/3.09 ms, NEX 1, slice thickness 5 mm with no gap, FOV 188 × 180 mm, matrix size 160 × 160, voxel size 0.625 × 0.625 × 5 mm3, time resolution 15.4 s Contrast agent administration: 0.05 mmol/kg, 3 mL/s, Gadobenate dimeglumine or Gadoterate meglumine Pharmacokinetic model: Patlak model.	No specific values provided.
Montagne et al. (2020)	196 NCI, 39 MCI	Scanner model: 3 T Siemens Trio; Coil configuration: 12-channel head coil	DCE-MRI scanning parameters: T1-weighted 3D VIBE pulse sequence, FA 15° TR/TE 5.14/2.18 ms, NEX 1, Slice thickness 5 mm with no gap, FOV 175 × 175 mm, Matrix size 320 × 320, Voxel size 0.550 × 0.550 × 5 mm3 time resolution 15.4 s. Contrast agent administration: Gadoterate meglumine 0.05 mmol/kg, 3 mL/s. Pharmacokinetic model: Patlak model.	Patients BBB Ktrans($\times 10^{-3} \text{ min}^{-1}$): APOE3: hippocampus = 1.29 (1.07–1.51), parahippocampal gyrus = 0.9 (0.79–1.04). APOE4: hippocampus = 1.59 (1.38–1.85), parahippocampal gyrus = 1.13 (0.97–1.26).
Freeze et al. (2020)	32 NCI, 34 MCI, 14 AD	Scanner model: 3.0 Tesla Philips Achieva	DCE-MRI scanning parameters: Dual resolution - fast scans (TR/TE 5.3/2.5ms, voxel	Patients BBB Ktrans($\times 10^{-3} \text{ min}^{-1}$): MCI: Total gray matter = 7.08, Total white matter

(continued on next page)

Table 1 (continued)

Study: author (year)	subject sample size	Scanner model and coil configuration	Imaging parameters	Main BBB findings
		Coil configuration: 32-channel head coil	2x2x5mm), then slow scans (TR/TE 5.6/2.5ms, voxel 1x1x2mm). Contrast agent administration: 0.1 mmol/kg, 3 mL/s, Gadobutrol injected during fast scans. Pharmacokinetic model: Patlak model.	= 18.6, Cortex = 8.5, Deep gray matter= 0.4, Hippocampus = 2.9. AD: Total gray matter = 7.4, Total white matter= 8.1, Cortex = 7.4, Deep gray matter = 0.8, Hippocampus = 4.5.
Choi et al. (2022)	23 Subjective cognitive impairment, 46 Early MCI, 35 Late MCI, 28 AD	MRI scanner: 3.0 T Siemens Magnetom Skyra Coil configuration: 20-channel head coil	DCE-MRI scanning parameters: Coronal sequence, Voxel size: 1.25 × 1.25 × 3 mm 60 dynamics, Time resolution: 10 s Contrast agent administration: Standard dose of gadobutrol Pharmacokinetic model: Patlak model.	Patients choroid plexus Ktrans(x 10 ⁻³ min ⁻¹): Subjective cognitive impairment = 0.56 (0.31–1.65) Early MCI= 0.66 (0.18–1.26) Late MCI = 1.16 (0.41–2.33) AD= 0.37 (0.18–1.12)
Moon et al. (2023)	14 MCI, 48 AD	MRI scanner: 3.0 Tesla Magnetom Skyra unit Coil configuration: 20-channel high-resolution head coil	DCE-MRI scanning parameters: TR: 3.1 s, TE: 1.04 ms, Flip Angle: 10 °, FOV: 225 × 240 mm, Slice Thickness: 3 mm, Matrix: 180 × 192, Voxel Size: 1.25 × 1.25 × 3 mm. Contrast agent administration: Gadobutrol 1.0 mmol/kg. Pharmacokinetic model: Patlak model.	Patients BBB Ktrans(x 10 ⁻³ min ⁻¹): Cortical: amyloid PET positive group= 0.9 ± 0.7 amyloid PET negative group= 0.8 ± 0.8 Hippocampal: amyloid PET positive group= 0.6 ± 0.7 amyloid PET negative group= 0.5 ± 0.5
Preis et al. (2024)	44 NC, 21 MCI, 26 AD	3 T Prisma fit MRI scanner Coil configuration: 64-channel head coil	DCE-MRI scanning parameters: T1-weighted images, TE= 2.5 ms, TR= 50 ms, Flip angle 60°, FOV 220, Voxel size 1.7 × 1.7 × 2.5 mm, 9 slices. Contrast agent administration: 10 mL Gadovist at a flow rate of 1 mL/s. Pharmacokinetic model: Patlak model.	Patients BBB Ktrans(x 10 ⁻³ min ⁻¹): Hippocampal: NC= 0.19 ± 0.37 MCI= 0.17 ± 0.22 AD= 0.55 ± 0.74
Reas et al. (2024)	44 CN, 11 cognitive impairment	3.0 Tesla Discovery 750 scanner	DCE-MRI scanning parameters: TR ranged from 7.7 to 8.2 ms, Flip Angle= 30 °, FOV= 256 × 256 mm, Slice Thickness= 5 mm, Slices = 28. Contrast agent administration: Gadobutrol, 0.1 mL/kg at a flow rate of 2 mL/s. Pharmacokinetic model: Patlak model.	Patients BBB Ktrans(x 10 ⁻³ min ⁻¹): CN: Gray matter = 0.90, White matter = 0.39, Hippocampus = 0.51, Entorhinal= 1.82, Parahippocampal = 1.12. Cognitive impairment: Gray matter = 0.93, White matter = 0.42, Hippocampus = 0.46, Entorhinal= 1.73, Parahippocampal = 1.28.

NCI = individual with no cognitive impairment, MCI = mild cognitive impairment, AD=Alzheimer’s disease, DCE-MRI = Dynamic Contrast-Enhanced Magnetic Resonance Imaging, BBB=Blood-Brain Barrier, APOE = Apolipoprotein E, SPGR= Spoiled Gradient Recalled, FA= Flip Angle, TR= Repetition Time, TE= Echo Time, NEX= Number of Excitations, GRE= Gradient Recalled Echo.

concentration of contrast agent in arteries and capillaries is the same; (4) The concentration of contrast agent is uniform within each compartment; (5) The exchange rate of water and cell membranes between tissues is far greater than the difference in T1 relaxation rates; (6) The backflow of contrast agent from tissues into blood vessels can be ignored during the measurement period(Bai et al., 2020b; Sourbron and Buckley, 2011). Translating these principles to clinical insights, pioneering Patlak analyses of DCE-MRI identified hippocampal BBB breakdown preceding both atrophy and CSF biomarker alterations in prodromal AD stages (Montagne et al., 2015). However, later cross-sectional MRI evaluations involving 91 participants revealed a critical distinction: AD patients exhibited selective hippocampal BBB disruption absent in MCI cohorts (Preis et al., 2024). Expanding this spatial perspective, subsequent work demonstrated AD-associated global hypoperfusion coupled with extensive BBB leakage patterns that spatially correlated with perfusion deficits, thereby reinforcing BBB dysfunction’s mechanistic role (van de Haar et al., 2016b). Furthermore, accumulating evidence documents progressive BBB permeability increases across broader neuroanatomical regions as AD advances(van de Haar et al., 2016a; van de Haar et al., 2017), with compelling clinical correlations to accelerating cognitive decline (Montagne et al., 2019; van de Haar et al., 2017). Notably, emerging evidence highlights genetic modulation of vascular vulnerability, as multimodal MRI studies detect elevated cortical BBB permeability in APOE4 carriers during preclinical AD phases(Reas et al., 2024), suggesting genetic predisposition may accelerate BBB dysfunction prior to overt cognitive symptoms. Recent paradigm-shifting findings from large-scale studies establish BBB disruption in early cognitive impairment as an Aβ/tau-independent contributor to AD(Montagne et al., 2020; Nation et al., 2019), though regional heterogeneity exists in permeability correlations with amyloid and tau deposition(Montagne et al., 2020). Pathoheterogeneity-nuanced analyses reveal differential Ktrans-CSF biomarker relationships contingent on amyloid status:

amyloid-positive individuals exhibit inverse cortical Ktrans-Aβ40 associations alongside positive Aβ42/40 and hippocampal volume correlations, whereas amyloid-negative cohorts demonstrate positive Ktrans-t-tau relationships(Moon et al., 2023). Complementing these observations, a retrospective MRI analysis of 532 patients along the AD continuum identified choroid plexus volumetric increases coupled with permeability reductions in advanced stages, proposing BBB breakdown as a potential progression biomarker(Choi et al., 2022). Finally, cross-sectional data spanning normal aging to dementia implicate BBB dysfunction in small vessel disease exacerbation and processing speed deficits mediated through white matter hyperintensity pathways(Freeze et al., 2020).

The aforementioned analytical evidence collectively underscores the pivotal role of DCE-MRI in characterizing BBB dysfunction during AD progression, thereby establishing a crucial imaging framework for elucidating associated pathological mechanisms. Nevertheless, three critical limitations warrant systematic consideration. First and foremost, the quantitative validity of DCE-MRI measurements remains intrinsically dependent on kinetic modeling approaches, where variations in model selection and parameter configuration inevitably compromise result reproducibility and cross-study comparability(Armitage et al., 2011; Chagnot et al., 2021). This methodological constraint is further compounded by the technique’s current resolution limitations. Specifically, DCE-MRI lacks the spatial resolution necessary for cellular or subcellular-level quantification, a technological gap that hinders mechanistic interpretations. Moreover, the explicit correlation between imaging-derived metrics and underlying pathophysiological processes demands rigorous validation through multidisciplinary investigations. To advance clinical translation, three strategic research directions emerge as imperative. Primarily, standardized multicenter trials employing harmonized scanning protocols and unified analytical models should be prioritized to enhance the accuracy and reliability of

DCE-MRI quantification. Expanding from this foundation, longitudinal cohort studies tracking dynamic BBB permeability changes alongside disease progression could yield clinically actionable insights for personalized therapeutic strategies. Notably, integrative analysis combining DCE-MRI parameters with complementary biomarkers from peripheral blood and CSF opens new avenues for both mechanistic exploration and novel biomarker discovery (Lee and Funk, 2023). This multidimensional approach may potentially disentangle the complex interplay between vascular dysfunction and neurodegenerative cascades in AD pathogenesis.

4.1.2. Water-exchange MRI: probing transvascular water dynamics

The quantification of trans-BBB water exchange dynamics represents a critical methodological frontier in neuroimaging research. Three principal non-invasive MRI-based methodologies have emerged as the cornerstone techniques for probing this physiological phenomenon: ASL, contrast enhancement, and double diffusion encoding. These techniques exploit measurable alterations in fundamental magnetic resonance parameters (including but not limited to apparent diffusion coefficient, T1 relaxation time, and T2 relaxation time) that arise from differential water mobility between intravascular and extravascular compartments. The underlying physical principle governing these measurements requires sufficient biophysical contrast in either compartmental water residence times or permeability-surface area products to enable robust quantification. Contemporary implementations of these techniques employ sophisticated biophysical modeling approaches that account for confounding factors such as partial volume effects, vascular pulsatility artifacts, and compartmental exchange time heterogeneity. Particularly in the context of AD pathophysiology, these methods provide unique insights into early BBB dysfunction preceding macroscopic pathological changes. The overview of BBB studies using water-exchange MRI on subjects with the AD continuum is presented in Table 2.

ASL technology magnetically labels inflowing arterial water molecules by inverting magnetization. Labeled water molecules flow into the imaging plane within a period of time after labeling. The ASL signal is acquired by measuring the signal difference between the two images before and after labeling. Importantly, leveraging differences in relaxation times and diffusivity between intravascular and extravascular compartments alongside exchange of magnetization between labeled protons and bulk water spins, ASL can ascertain BBB water exchange rate and blood volume fraction (Detre et al., 2012; Dickie et al., 2020; Li et al., 2022; Shao et al., 2019). Notably, because T1 relaxation time differs minimally between blood and brain, some techniques combine standard ASL with other magnetization preparation to better assess BBB water exchange (Gregori et al., 2013; Schidlowski et al., 2020; Silva et al., 1997; Wang et al., 2012), including diffusion-weighting (Shao et al., 2019; Wang et al., 2006b) and multi-echo time ASL (Mahroo et al., 2021; Wells et al., 2009). Emerging evidence from clinical studies demonstrates the biomarker potential of this approach. One ASL study showed that BBB water exchange rate (Kw) positively correlated with CSF A β 42 levels across several brain regions, suggesting Kw may be a promising noninvasive marker of A β clearance capacity (Gold et al., 2021). However, conflicting findings have been reported in subsequent investigations. Another study found the opposite, with higher BBB permeability associating with elevated CSF p-tau and lower A β 42/A β 40 ratio (Lin et al., 2021) (Fig. 4). To further elucidate these relationships, an observational ASL and Pittsburgh Compound B-PET study revealed inverse correlations between Kw and SUVR in the frontal and medial temporal lobes, implying diminished BBB-mediated waste clearance leads to A β accumulation and cognitive decline in APOE4 carriers due to their low BBB water exchange (Uchida et al., 2022). Consistent with these observations, compared to controls, AD patients exhibited reduced perivascular clearance rate near the BBB by ASL in six brain regions (Joseph et al., 2020). The clinical relevance of BBB water exchange metrics extends beyond neurodegenerative diseases. One study linked

increased BBB water exchange to more vascular risk factors and poorer cognition (Shao et al., 2019). Specifically, individuals with hypertension, hyperlipidemia, diabetes or tobacco use showed lower cortical and white matter Kw than those without these factors (Ford et al., 2022). These findings collectively suggest that diminished Kw may reflect ongoing neurodegeneration. To validate the translational potential of these measurements, using multi-echo time ASL-MRI in mice, BBB water exchange time was significantly prolonged in AQP4 knockouts, demonstrating this technique can effectively evaluate blood-brain water transport (Ohene et al., 2019).

Transitioning from conventional methods, contrast-enhanced MRI techniques utilize gadolinium administration to shorten intravascular T1/T2 relaxation times, thereby amplifying the inherently subtle contrast between intra- and extravascular compartments (Dickie et al., 2021; Dickie et al., 2019; Rooney et al., 2015). Notably, this approach enables simultaneous quantification of both water exchange rates (reflecting per-molecule exchange frequency) and total trans-BBB water flux within defined temporal windows (Dickie et al., 2020). A pivotal demonstration of this methodology emerged from a DCE-MRI study revealing elevated BBB water permeability in AD model mice, which correlated with diminished TJ protein occludin expression (Dickie et al., 2019). Intriguingly, cumulative evidence suggests that both aging and AD pathology exacerbate BBB dysfunction, with AD progression accelerating permeability increases through distinct mechanistic pathways (Dickie et al., 2021).

In contrast to contrast-enhanced and ASL techniques, filter-exchange imaging eliminates the need for exogenous contrast agents or spin labeling by exploiting intrinsic differences in water diffusion characteristics between vascular and extravascular compartments (Lampinen et al., 2017; Lasić et al., 2011; Nilsson et al., 2013). Mechanistically sophisticated, this double-diffusion encoding strategy employs sequential diffusion weighting blocks: the first suppresses fast-diffusing vascular signals, followed by a controlled exchange period, and subsequent encoding of remaining spins through secondary diffusion weighting. Critically, the temporal evolution of apparent diffusion coefficient increments correlates with exchange duration, while recovery kinetics are governed by compartmental exchange rates. Despite its technical elegance, implementation challenges persist, particularly regarding compartmental relaxation rate disparities and longitudinal crusher gradient optimization (Bai et al., 2020a; Lasić et al., 2017; Motta et al., 2017). Clinically relevant advancements include recent adaptations for BBB water exchange assessment, with hippocampal permeability elevations detected in mild cognitive impairment patients versus controls – a phenomenon exacerbated in AD cases with progressive involvement of broader brain regions (Zhang et al., 2023).

Collectively, these emerging MRI-based methodologies for quantifying trans-BBB water exchange not only demonstrate substantial potential for noninvasive evaluation of BBB integrity alterations associated with AD pathogenesis, but also provide mechanistic insights into functional changes at the blood-brain interface (Elschot et al., 2021). A principal strength of these techniques lies in their capacity to circumvent the need for gadolinium-based contrast agent administration, thereby eliminating associated safety concerns and simplifying clinical protocols. Nevertheless, it is important to acknowledge that current biophysical models face inherent challenges in precisely differentiating compartment-specific signals, primarily due to the intricate nature of water exchange dynamics across the BBB. To address these limitations, the field is currently progressing along two complementary axes: (1) systematic optimization of acquisition protocols and analytical frameworks to account for compartmental disparities in relaxivity/diffusivity profiles and (2) development of advanced modeling approaches capable of disentangling contributions from multiple exchange pathways. Particularly noteworthy is the critical need for next-generation exchange models that can discriminate between permeability-mediated transport and active transporter-regulated water flux—a distinction paramount for understanding AD-related BBB pathophysiology.

Table 2

Overview of studies using water-exchange MRI and DGE-MRI to assess blood-brain barrier permeability in AD continuum.

Study: author (year)	subject sample size	MRI scanner and coil configuration	Imaging parameters	Main BBB findings
Arterial spin labeling MRI				
Shao et al. (2019)	10 NCI, 9 MCI	Scanner model: Siemens 3 T Prisma Coil configuration: 20-channel head coil	DP-pCASL parameter: Field of view: 224 mm, Matrix size: 64 × 64, Slice thickness: 8 mm, Number of slices: 12 slices (10 % oversampling), Resolution: 3.5 × 3.5 × 8 mm ³ , TE: 36.5 ms, TR: 4000 ms, Labeling duration: 1500 ms, ATT measurement: PLD 900 ms, b-values: 0 s/mm ² , 10 s/mm ² (VENC) 15 repetitions each; kw measurement: PLD 1800 ms, b-values: 0 s/mm ² , 50 s/mm ² , 20 repetitions each. Water exchange model: Regularized SPA.	Subjects BBB kw value (min ⁻¹): Whole brain= 105.0 ± 20.6 Gray matter= 109.6 ± 18.9 White matter= 94.1 ± 19.6
Ohene et al. (2019)	5 Wild type C57BL/6 mice, 9 Aqp4 knockout mice, 9 Wild type controls	Scanner model: 9.4 T VNMRs horizontal bore Coil configuration: 72 mm volume coil, Two channel array surface coil	Multi-TE ASL parameters: TE: 15, 23, 30, 40, 50, 65 ms, TR: 5000 ms (except 3500 ms TI with TR 7000 ms), FOV: 25 × 25 mm, Matrix size: 32 × 32, Slice thickness: 2 mm. Exchange model: Two compartment kinetic perfusion model	Subjects BBB exchange time value (ms): Wild type cortical = 343 ± 91 Aqp4-/- cortical = 452 ± 90
Joseph et al. (2020)	18 NCI, 3 mild AD	Scanner model: Siemens 3 T Skyra Coil configuration: 20 channel head coil or flexible 4 channel coil	3D TGSE PASL parameters: TI: 7 inversion times from 2800 ms to 4000 ms in 200 ms increments, TR: Varies from 3830 ms to 5320 ms depending on TI, TE: 16.36 ms, Matrix: 63 × 64 × 40, Voxel size: 3.9 × 3.9 × 4 mm ³ , Labeling duration: 700 ms, Slices: 40 contiguous 4 mm slices. Water exchange model: T1 Signal Decay Patterns	No specific values provided.
Gold et al. (2021)	39 preclinical AD	Scanner model: 3 T Siemens PRISMA Coil configuration: 64-channel head coil	DP-pCASL parameter: TR = 4 sec, TE = 36.5 ms, FOV = 224 mm, matrix size = 64 × 64, 12 slices (10 % oversampling), resolution = 3.5 × 3.5 × 8 mm ³ , label/control duration = 1500 ms, PLD= 1800ms, b values = 0, 50 s/mm ² , repetitions= 20, acquisition time= 6 min. Water exchange model: TGV Regularized SPA	Subjects BBB kw Values (min ⁻¹): Whole brain= 98.27 ± 19.77 Frontal lobe= 104.42 ± 22.18 Parietal lobe= 83.21 ± 28.34 Precuneus= 85.96 ± 28.79 Medial temporal lobe= 85.87 ± 27.32
Lin et al. (2021)	33 MCI, 22 NCI	Scanner model: 3 T Philips scanner Coil configuration: Head coil	WEPCAST sequence parameters: FOV = 200 × 200 mm ² Matrix= 64 × 64, Voxel size = 3.13 × 3.13 mm ² , Slice thickness= 10 mm, FA= 90 °, TR= 7547 ms, TE= 13 ms, Number of control/label pairs= 10. Water exchange model: non-single pass model.	Subjects BBB PS Values (mL/100 g/min): MCI= 142.6 ± 25.6 Control= 123.0 ± 26.0
Uchida et al. (2022)	66 early AD (24 APOE ε4 noncarriers, 22 heterozygotes, 20 homozygotes)	Scanner model: 3.0 T Philips Ingenia Coil configuration: 32-channel head coil	DP-pCASL parameters: TR 4100 ms, TE 10.3 ms, FOV 224 mm, matrix 64 × 64, slices 42, Resolution 3.5 × 3.5 × 7mm ³ , labelled duration 1800 ms Five repetitions per control/label, ATT from b= 13.4 s/mm ² , PLD= 900ms; kw from b= 50 s/mm ² , PLD= 1800ms. Water exchange model: Regularized SPA.	Subjects BBB kw value (min ⁻¹): Frontal lobe: APOE ε3/ε4 = 94.11 ± 8.34, APOE ε4/ε4 = 77.95 ± 6.86, Medial temporal lobe: APOE ε3/ε4 = 86.73 ± 8.12, APOE ε4/ε4 = 76.08 ± 6.95, Lateral temporal lobe: APOE ε3/ε4 = 93.35 ± 8.07, APOE ε4/ε4 = 87.36 ± 7.67, Parietal lobe: APOE ε3/ε4 = 92.56 ± 8.01, APOE ε4/ε4 = 88.82 ± 7.14 Precuneus: APOE ε3/ε4 = 86.80 ± 7.29, APOE ε4/ε4 = 81.73 ± 6.91, Posterior cingulate gyrus: APOE ε3/ε4 = 84.04 ± 7.11, APOE ε4/ε4 = 81.06 ± 7.23.
Ford et al. (2022)	26 NCI, 4 MCI	Scanner model: 3.0 T GE Discovery Coil configuration: 32-channel head coil	DP-pCASL with QPM parameter: voxel size = 1.9 × 1.9 × 4 mm ³ , Spiral readouts for off-resonance robustness at 3 T, Adiabatic BIR-4 diffusion preparation, Multiple b-values: 0, 10, 20, 50, 100 s/mm ² , Multiple PLDs: 1000, 1500, 1800, 2000, 2500 ms, Total scan time: 27 min.	Subjects BBB kw value (min ⁻¹): Cortex = 81.5 ± 15.4 white matter = 75.2 ± 13.9
Double diffusion encoding MRI				
Zhang et al. (2023)	27 NCI, 14 MCI, 11 AD	Scanner model: 3.0 T Siemens Prisma Coil configuration: 64-channel head coil	Vascular-water-exchange MRI parameters: Voxel size: 3.0 × 3.0 × 5.0 mm ³ , Number of slices: 20, Filter block: PGSE, 90° pulse, b= 250 s/mm ² gradient 180° pulse, TE= 26ms. Mixing time: 25/200/400ms dephasing. Detection block: 90° pulse, b= 0,250 s/mm ² gradients, 180°	No specific values provided.

(continued on next page)

Table 2 (continued)

Study: author (year)	subject sample size	MRI scanner and coil configuration	Imaging parameters	Main BBB findings
			pulse, TE= 37ms. Post-processing: Signal fitting, Extract Kw.	
Contrast enhancement MRI Dickie et al. (2019)	7 transgenic TgF344-AD rats, 5 wild-type littermates	Scanner model: Bruker Avance III 7 T Coil configuration: Rat brain surface coil	Multi-flip angle multi-echo SPGR parameters: Orientation: Three-dimensional, Repetition time (TR): 100 ms, Number of flip angles: 5, Flip angle values: 10, 20, 30, 40, 80 °, Number of gradient echoes: 10 Matrix size: 32 × 16 × 32, Number of slices: 96, Number of repetitions: 6. Water exchange model: Two-site one-exchange model	No specific values provided.
Dickie et al. (2021)	13 months: 13 TgF344-AD, 16 wildtype 18 months: 7 TgF344-AD, 5 wildtype 21 months: 8 TgF344-AD, 7 wildtype	Scanner model: 7 T Bruker Coil configuration: T11070V3 transceiver coil, T11205V3 surface receiver coil	Multi-flip angle multi-echo SPGR sequence: TR: 100 ms, TE: 2.14 ms, Echo spacing: 2.14 ms, Number of echoes: 10, Flip angles (α): 30°, 40°, 20°, 10°, 80°, 3D acquisition, Slice thickness: 1 mm, In-plane resolution: 1 × 1 mm ² , Matrix size: 32 × 32 × 30. Water exchange model: Two-site water exchange model.	Subjects BBB P _{Sw} value (mL/ min/mL): 13 months: TgF344-AD= 0.040 ± 0.010, Wildtype= 0.035 ± 0.007 18 months: TgF344-AD= 0.057 ± 0.018, Wildtype= 0.033 ± 0.004 21 months:TgF344-AD= 0.059 ± 0.021, Wildtype= 0.074 ± 0.022
Dynamic glucose-enhanced MRI Huang et al. (2020)	10 Wild-type mice, 10 APP/ PS1 transgenic AD mice	Scanner model: 3 T animal Coil configuration: Surface coil	D-glucose MRI parameters: TR/TE: 200/5 ms, FOV: 20 × 20 mm ² , Matrix: 128 × 128, Slice thickness: 0.5 mm, Temporal resolution: 15 s Acquisition time: 60 min.D -glucose administration: D-glucose injected via tail vein at beginning, maintained at 20–30 mM	Subjects D-glucose uptake: (Signal changes relative to the baseline) Younger AD mice parenchyma= 2.16 ± 0.38 % Older AD mice parenchyma= 0.87 ± 0.09 % (Dynamic change rate constant) Younger AD mice CSF= 0.09 ± 0.03 min ⁻¹
Chen et al., (2021)	4 Tau4RDK mice, 4 C57BL/6 J wild-type mice	Scanner model: 11.7 T Bruker Biospec Coil configuration: mouse brain phased array receiver coil	D-glucose MRI parameters: Binomial pulses: 12 pulses (36 ms), 100 pulses (300 ms), TR/TE: 2.5 s/5ms,FOV: 20x10mm ² , Matrix: 196 × 128 Slice thickness: 1 mm, Dynamic scans: 35 min, 10 min pre- infusion.D -glucose administration: 0.15 mL 50 % dextrose over 1 min	Subjects D-glucose uptake: (Signal changes relative to the baseline) Tau mice cortex= 1.70 % ± 1.33 % WT mice cortex= 5.42 % ± 0.70 % (Dynamic change rate constant) Tau mice CSF= 0.08 ± 0.01 min ⁻¹ WT mice CSF= 0.56 ± 0.1 min ⁻¹

NCI = individual with no cognitive impairment,MCI = mild cognitive impairment, AD=Alzheimer’s disease, BBB=Blood-Brain Barrier, APOE = Apolipoprotein E, FA= Flip Angle, TR= Repetition Time, TE= Echo Time, NEX= Number of Excitations, GRASE=gradient and spin-echo, QPM=quantitative permeability mapping, SPA = single-pass approximation, TGSE=turbo gradient spin echo, DP-pCASL=diffusion-prepared pseudo-continuous arterial spin labeling, PASL=pulsed arterial spin labeling, FOV = Field of view

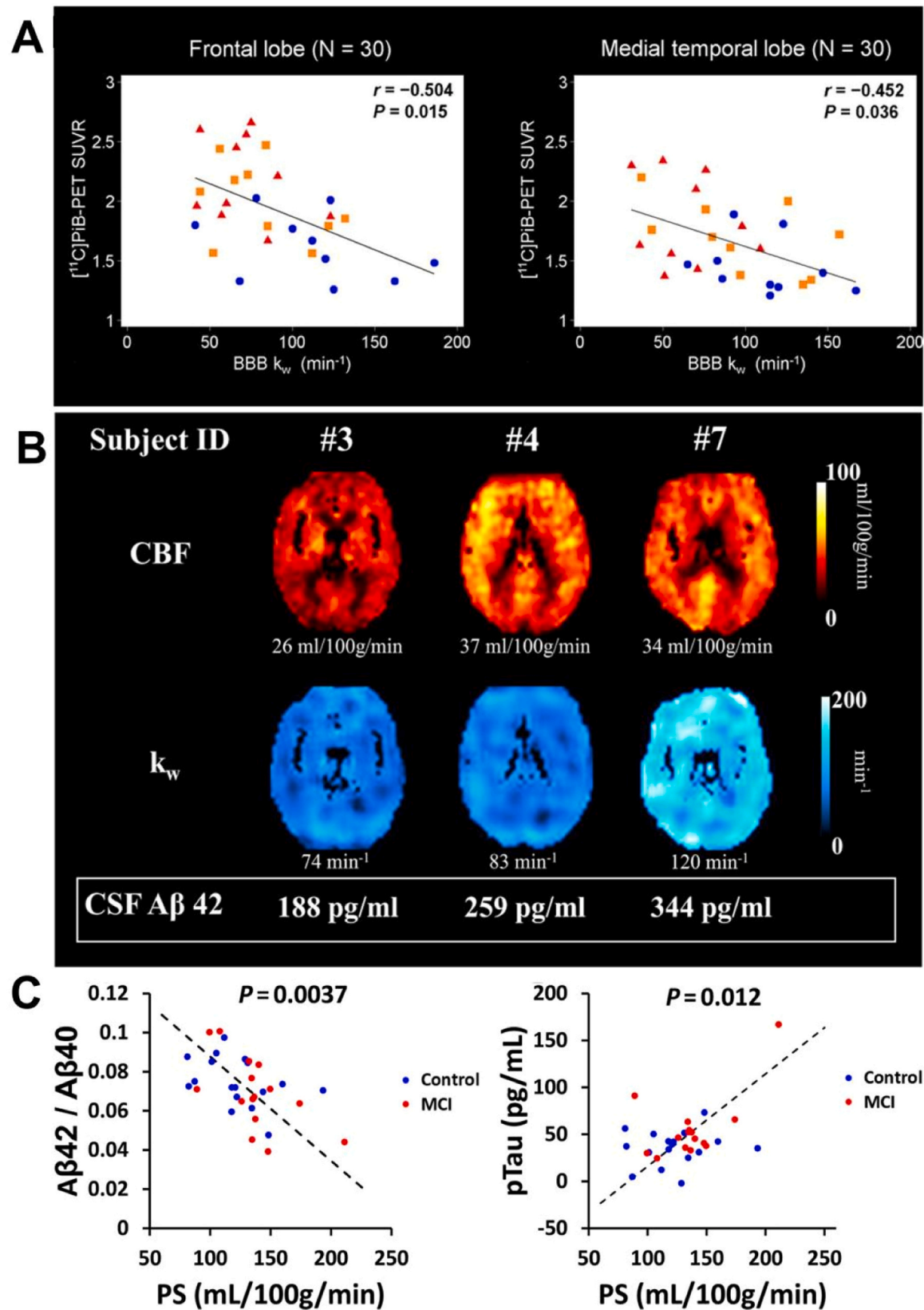
Building upon these considerations, ongoing validation studies correlating water exchange MRI metrics with established BBB integrity markers will be essential for establishing biological relevance and clinical utility. With continued methodological refinement and rigorous cross-modal validation, this technique may ultimately provide a sensitive, mechanism-based tool for mapping BBB dysfunction trajectories in AD, particularly through its unique ability to noninvasively probe specific water transport pathways across the neurovascular unit.

4.1.3. DGE-MRI: mapping glucose transport deficits

DGE-MRI represents a groundbreaking advancement in molecular imaging that harnesses endogenous D-glucose as a biocompatible contrast agent (Fig. 5). This technique capitalizes on the CEST mechanism, where frequency-selective radiofrequency pulses saturate hydroxyl protons on circulating D-glucose molecules. Through sophisticated pulse sequence design, the system dynamically quantifies CEST signal variations arising from proton exchange between these saturated glucose protons and bulk water protons, enabling real-time metabolic imaging without exogenous contrast administration(Paech and Radbruch, 2021; Paech et al., 2017; Xu et al., 2015a). The molecular

architecture of D-glucose, featuring five hydroxyl groups, confers exceptional CEST sensitivity even at physiological concentrations (4–6 mM in brain tissue). This intrinsic molecular property eliminates contrast agent toxicity concerns while maintaining sufficient signal-to-noise ratios for quantitative imaging(Paech and Radbruch, 2021; Paech et al., 2017; Xu et al., 2015a). When applied to neuroimaging, DGE-MRI provides unprecedented spatial-temporal resolution for mapping cerebral glucose influx kinetics, establishing itself as a transformative tool for noninvasive evaluation of BBB glucose transporter dysregulation in vivo(Huang et al., 2020). Advanced compartmental modeling algorithms further enhance the technique’s diagnostic potential by resolving two critical physiological parameters: (1) Transmembrane transport constants reflecting GLUT density and activity; (2) Metabolic rate constants indicative of intracellular glucose utilization. These quantitative biomarkers enable precise differentiation of glucose transporter subtypes and their functional contributions to cerebral metabolic homeostasis(Dickie et al., 2022). The overview of BBB studies using DGE-MRI on subjects with the AD continuum is presented in Table 2.

A investigation demonstrated markedly reduced D-glucose uptake in



(caption on next page)

Fig. 4. Pathophysiological implications of blood-brain barrier water exchange dynamics. The precise pathophysiological interpretation of kw across the BBB remains a subject of ongoing scientific debate. Previous multiple studies reveal two distinct mechanistic interpretations supported by neuroimaging and biomarker evidence: (A) Enhanced clearance hypothesis: Regional elevation of BBB kw in the frontal and medial temporal lobes exhibits a significant inverse correlation with amyloid- β deposition as quantified by [11 C]PIB-PET SUVR (Obtained from Uchida et al. (Uchida et al., 2022)). This spatial correspondence suggests that increased water flux may facilitate amyloid clearance through perivascular drainage pathways. (B) CSF biomarker correlation: Whole-brain BBB kw measurements demonstrate a positive association with cerebrospinal fluid A β 42 concentrations (Obtained from Gold et al. (Gold et al., 2021)). This finding reinforces the clearance hypothesis while indicating potential systemic differences in amyloid homeostasis between brain compartments. (C) Barrier integrity hypothesis: Contrasting these observations, the PS of water shows an inverse relationship with CSF A β 42/A β 40 ratio (Obtained from Lin et al. (Lin et al., 2021)), coupled with positive correlations with ptau levels. These divergent results highlight the complexity of BBB dynamics interpretation, where methodological differences in parameter measurement and regional versus global analyses may account for apparent discrepancies. The current evidence base supports two non-exclusive paradigms: 1) water exchange increases reflecting active clearance mechanisms in early-stage pathology, and 2) water exchange elevations indicating progressive barrier deterioration in advanced disease states. PET = positron emission tomography, BBB = blood-brain barrier, CSF = cerebrospinal fluid, A β 42 = amyloid-beta 42, ptau = phosphorylated tau, SUVR = standardized uptake value ratio, PS = permeability-surface area product, kw = water exchange rate.

both cerebral cortical tissue and CSF of AD mouse models compared to wild-type counterparts, mechanistically linked to dual transporter impairments at the BBB and blood-CSF barrier interfaces (Chen et al., 2021). Notably, subsequent analysis revealed concomitant deterioration of CSF glucose clearance kinetics in AD mice, implicating parallel dysfunction in glymphatic drainage pathways (Chen et al., 2021). Intriguingly, while corroborating these fundamental observations in brain parenchyma and CSF compartments, a separate study identified a paradoxical age-dependent reversal – juvenile AD mice exhibited elevated parenchymal glucose accumulation relative to age-matched controls, a phenomenon hypothetically attributed to developmental-stage metabolic compensation preceding age-associated transporter decline (Huang et al., 2020). In this technological context, DGE-MRI has emerged as a transformative neuroimaging modality enabling non-invasive interrogation of BBB integrity and associated metabolic perturbations (Huang et al., 2020). Particularly advantageous over conventional nuclear imaging techniques (e.g., PET, SPECT), DGE-MRI provides superior spatial resolution essential for quantifying dynamic neurovascular coupling phenomena (Tolomeo et al., 2018). Nevertheless, several critical limitations constrain current applications. Foremost, the requisite supra-physiological glucose loads for signal detection may confound interpretation by transiently modulating transporter expression profiles and barrier permeability (Knutsson et al., 2018). Equally problematic, the absence of standardized methodologies for deriving reliable arterial input functions fundamentally limits pharmacokinetic modeling precision (Knutsson et al., 2018). Furthermore, the technique's specificity for particular transporter subtypes and capacity to differentiate distinct BBB pathological states remain inadequately validated through targeted quantitative analyses (Jin et al., 2018). Looking forward, strategic development of BBB-penetrant glucose tracers (including optimized FDG derivatives, non-metabolizable analogs, and 3-O-[11 C]-methylglucose formulations) could revolutionize spatial mapping of trans-barrier glucose flux. Concurrently, systematic implementation of standardized preclinical protocols across diverse AD animal models will be imperative to comprehensively evaluate DGE-MRI's potential for elucidating BBB pathomechanisms in evolving AD pathogenesis.

4.2. PET imaging

PET has emerged as a cutting-edge molecular imaging modality for evaluating BBB dysfunction in AD, leveraging its unique capability to quantitatively map neurovascular pathophysiology at molecular scales (Fig. 6). This nuclear imaging technique employs ultra-sensitive radiation detectors coupled with radiochemically engineered biomarkers to achieve picomolar-level detection sensitivity, enabling dynamic monitoring of BBB transport mechanisms through kinetic modeling. Unlike relative quantification methods, PET provides absolute quantitation of tracer pharmacokinetics through arterial input function analysis and compartmental modeling, allowing precise characterization of transporter expression and functional activity. The armamentarium of BBB-targeted PET radiopharmaceuticals has expanded significantly in AD

research. [18 F]-FDG-PET serves as a paradigmatic example, quantifying glucose transport kinetics via GLUT1 transporters and revealing region-specific BBB glucose hypometabolism in early AD stages. Equally notable, [11 C]-verapamil PET has uncovered compromised P-gp-mediated amyloid-beta clearance through multi-timepoint graphical analysis. Second-generation tracers now target diverse BBB components including AQP4 water channels (e.g., [18 F]-TGN-020), LRP1 receptors ([11 C]-LY-411575), and matrix metalloproteinases ([18 F]-BR-351), enabling multiparametric assessment of BBB dysregulation. This molecular imaging revolution provides unprecedented insights into spatiotemporal progression of BBB pathology across the AD continuum and new capabilities to quantitatively analyze BBB molecular alterations contributing to AD onset and progression. The overview of BBB studies using PET on subjects with the AD continuum is presented in Table 3.

4.2.1. FDG-PET: GLUT1 dysfunction and metabolic decline

As a glucose analog, FDG crosses the BBB through identical transport mechanisms to glucose. Notably, following phosphorylation by hexokinase into FDG-6P, it escapes further metabolic processing and becomes trapped intracellularly. This unique biochemical property enables PET imaging with FDG to quantitatively estimate the cerebral metabolic rate of glucose (Bailly et al., 2015; Iaccarino et al., 2017; Mosconi, 2005; Mosconi et al., 2013; Mosconi et al., 2008; Protas et al., 2013). Of particular clinical relevance, impaired early brain FDG uptake has been consistently observed in APOE4 carriers (Protas et al., 2013), PSEN1 mutation carriers (Bateman et al., 2012; Mosconi et al., 2006), and individuals with maternal AD history (Mosconi et al., 2013)—findings that collectively suggest compromised glucose delivery precedes substantial neuronal metabolic alterations in genetically susceptible populations. Longitudinal evidence consolidation demonstrates progressive FDG uptake reduction paralleling cognitive decline and neuropathological progression, ultimately culminating in AD manifestation (Bouter et al., 2019; Iaccarino et al., 2017; Landau et al., 2011; Mosconi et al., 2009; Ou et al., 2019). Beyond diagnostic implications, the FDG-PET-derived rate constant K1 serves as a specific biomarker for GLUT1-mediated BBB glucose uptake, independent of plasma glucose concentration. This functionality establishes FDG-PET as a unique modality capable of simultaneously evaluating both cerebral glucose metabolism and quantifying BBB glucose transport efficiency (Piert et al., 1996). Recent therapeutic investigations employing kinetic FDG-PET modeling revealed that 6-month liraglutide treatment significantly enhanced estimated maximum BBB glucose transfer capacity, contrasting with placebo-treated subjects who exhibited progressive decline correlating with AD progression (Gejl et al., 2017). This finding provides the first in vivo evidence of pharmacologically enhanced BBB glucose transport in AD patients. Expanding the scope of investigation, multiple kinetic FDG-PET studies have systematically characterized BBB glucose transport abnormalities in AD. Consistently observed patterns include reduced cortical influx constant K1 values (Piert et al., 1996), with particular vulnerability in frontal, temporal and parietal regions compared to healthy controls (Jagust et al., 1991). Intriguingly, regional analyses revealed most pronounced K1 reductions in the hippocampus

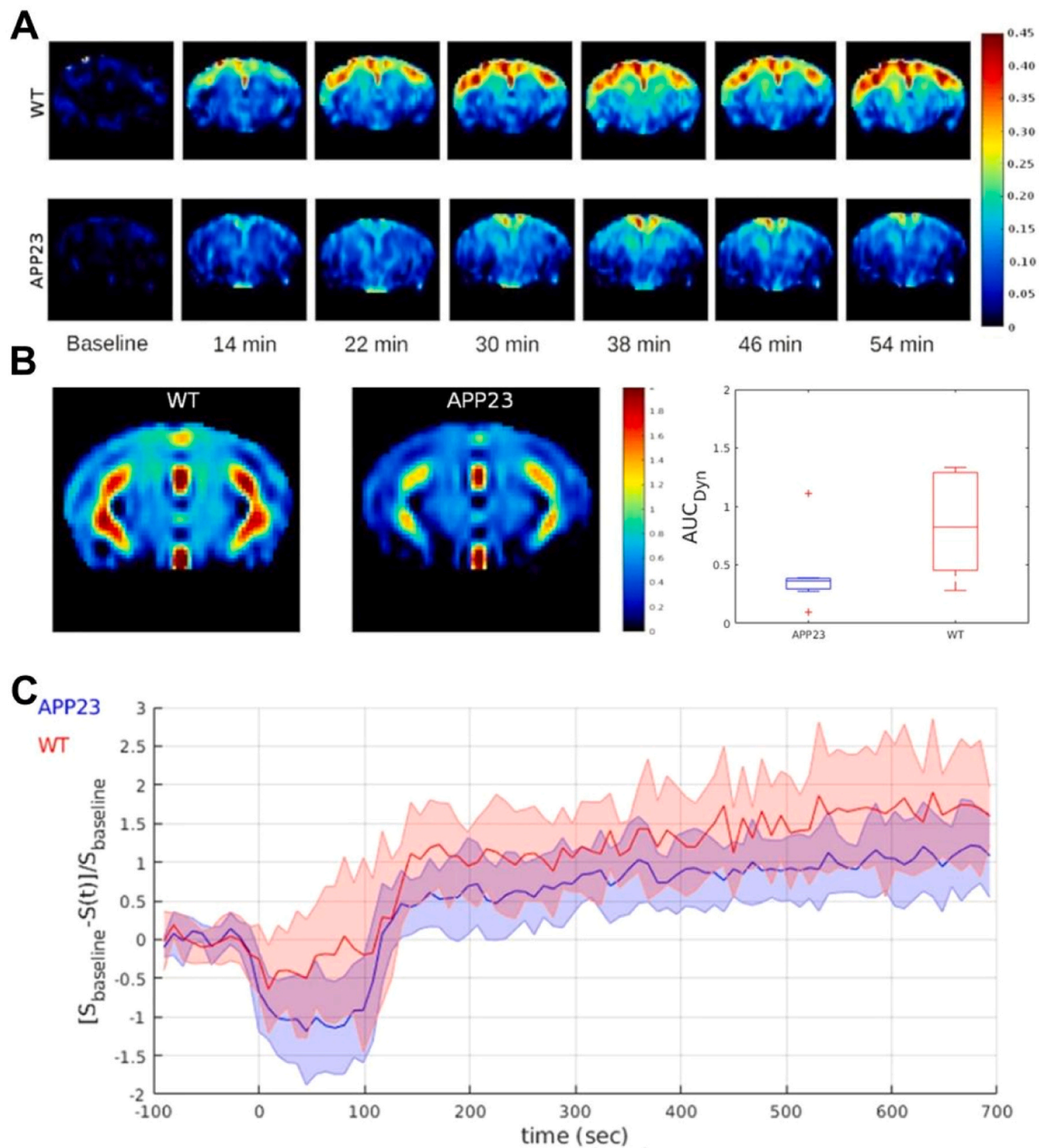


Fig. 5. A. The temporal evolution of dynamic CEST measurements is graphically delineated through longitudinal imaging analysis. WT murine models demonstrate persistently elevated glucose uptake kinetics compared to APP23 transgenic counterparts across all observation phases (Baseline–54 min post-injection), indicative of preserved metabolic functionality. B. Spatially normalized group-averaged dynamic enhancement maps illustrate neurovascular coupling patterns, registered to a standardized anatomical template through cortical surface area-based voxelwise computation. Quantitative pharmacokinetic analysis reveals substantial between-group differences in contrast agent distribution, with WT specimens exhibiting greater AUC values relative to APP23 models. C. Cortical signal-averaged dynamic enhancement curves demonstrate genotype-dependent metabolic trajectories. The temporal resolution of these measurements enables precise characterization of regional cerebral metabolic flux variations associated with amyloid pathology progression (Obtained from Tolomeo et al. (Tolomeo et al., 2018)). CEST = chemical exchange saturation transfer, AUC = area under the curve, WT = wild-type, APP23 = transgenic Alzheimer's disease model.

along with temporoparietal cortices (Mosconi et al., 2007), topographically aligning with characteristic AD neuropathology and suggesting region-specific BBB transport impairment. Mechanistic validation comes from transgenic models combining A β precursor protein overexpression with hemizygous GLUT1 deletion, which recapitulate key features of human AD including diminished cerebral 2-deoxyglucose uptake, subsequent BBB dysfunction, hypoperfusion, and progressive neurodegeneration (Winkler et al., 2015). Collectively, this converging evidence from multimodal human studies and experimental models establishes BBB glucose transport deficiency as a pathogenic contributor to AD progression, thereby highlighting the potential of FDG-PET

biomarkers for both early detection and therapeutic monitoring in neurodegenerative disorders.

4.2.2. Innovations in PET using other tracers: Targeting P-gp, RAGE, and AQP4

The functional assessment of P-gp, a critical efflux transporter at the BBB, has been significantly advanced through PET tracers such as [11 C]-verapamil, [11 C]-N-desmethyl-loperamide, and [11 C]-metoclopramide, all of which have received clinical approval for evaluating P-gp transport activity (Marie and Tournier, 2020). Notably, impaired amyloid clearance in AD has been mechanistically linked to P-gp

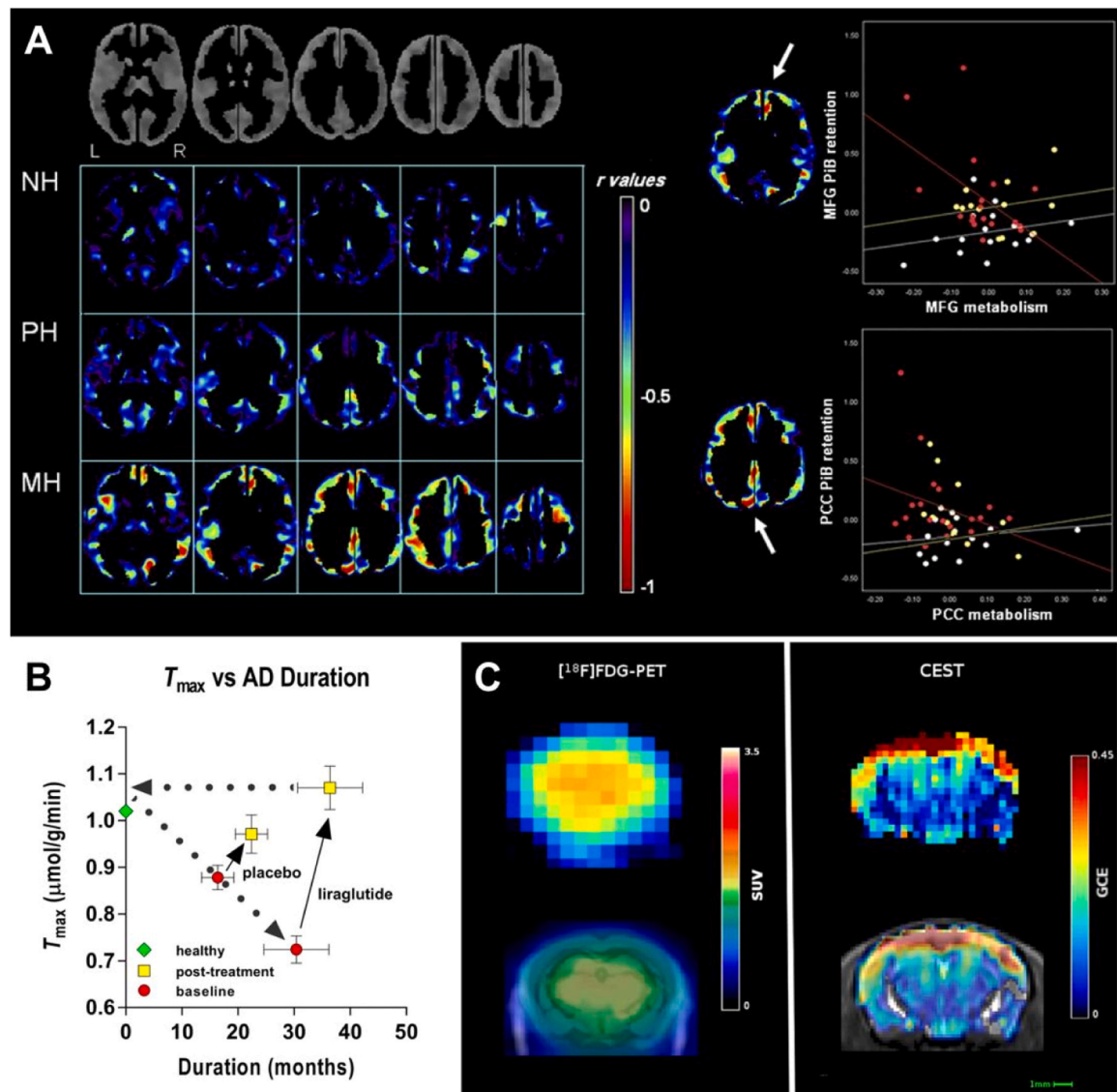


Fig. 6. A. Regional neuroimaging associations between cerebral amyloid- β deposition (Pittsburgh compound B retention) and cerebral metabolic rate of glucose utilization in cognitively intact subjects stratified by familial Alzheimer's disease risk. The cohort was categorized into three familial risk subgroups: NH, PH, MH. The results showed that compared to NH and PH individuals, MH individuals exhibited a tendency towards negative correlation between PIB retention and glucose metabolism. PET data processing incorporated gray matter segmentation and spatial normalization, with axial projections illustrating differential spatial patterns. Subsequent region-of-interest analysis identified significant negative correlations between medial frontal gyrus amyloid burden and posterior cingulate cortex hypometabolism exclusively in MH carriers, suggesting maternal-specific transmission of endophenotypic vulnerability (Obtained from Mosconi et al. (Mosconi et al., 2013)). B. Therapeutic potential of glucagon-like peptide-1 receptor agonists in Alzheimer's disease pathophysiology through blood-brain barrier modulation. Longitudinal assessment of blood-to-brain transfer constants (K_i values) demonstrated differential trajectories between treatment arms: placebo subjects exhibited progressive decline in glucose transporter type 1-mediated transport capacity, whereas liraglutide-treated patients maintained blood-brain barrier integrity. This pharmacodynamic response suggests GLP-1 analogs may counteract neurovascular uncoupling through GLUT1 stabilization, potentially attenuating bioenergetic failure in Alzheimer's progression (Obtained from Gejl et al. (Gejl et al., 2017)). C. Comparative neuroimaging characteristics of positron emission tomography in cerebral glucose transport assessment. While PET enables absolute quantification of tracer kinetics (K_1 : influx constant; k_2 : efflux constant) with sensitivity to nanomolar concentration changes, its spatial resolution (typically 4–5 mm full width at half maximum) imposes limitations in resolving small cerebral structures compared to high-field MRI (submillimeter isotropic resolution). This technical constraint necessitates careful interpretation of partial volume effects in regions with intricate cytoarchitecture (Obtained from Tolomeo et al. (Tolomeo et al., 2018)). PET = positron emission tomography, PIB = Pittsburgh compound B, FDG = fluorodeoxyglucose, GLP-1 = glucagon-like peptide-1, GLUT1 = glucose transporter type 1, MRI = magnetic resonance imaging, NH = no parental history of dementia, PH = paternal lineage history, and MH = maternal lineage history.

dysfunction. Reduced P-gp expression/function in AD pathophysiology correlates with compromised A β clearance (Storelli et al., 2020). [11 C]-verapamil-PET studies consistently reveal diminished BBB P-gp activity across multiple cortical regions in AD patients, including the frontal, parietal, temporal, and occipital cortices, as well as the posterior and anterior cingulate gyri (van Assema et al., 2012). Importantly, a separate [11 C]-verapamil-PET investigation demonstrated significant

P-gp functional reduction in mild AD patients, affecting the frontal, parietotemporal, and cingulate cortices along with the hippocampus—independent of perfusion deficits (Deo et al., 2014). While [11 C]-verapamil exhibits commendable selectivity and reproducibility for P-gp assessment, its clinical utility is constrained by high ligand affinity and susceptibility to partial volume effects. To circumvent these limitations, next-generation tracer development has adopted dual

Table 3

Overview of studies using PET to assess blood-brain barrier permeability in AD continuum.

Study:author (year)	subject sample size	PET scanner and coil configuration	Imaging parameters	Main BBB findings
[18 F]-fluorodeoxyglucose -PET Jagust et al. (1991)	6 AD, 7 NCI	PET scanner: Donner 600-crystal tomograph.	Kinetic FDG-PET parameters: Resolution: 2.6 mm in-plane FWHM, 6 mm axially Scan: Dynamic list mode acquisition over 40 min Frame durations: 5 s for first minute, 10 s from 1 to 2 min, 30 s from 2 to 4 min, 60 s from 4 to 10 min, 300 s from 10 to 40 min. Radiotracer administration: 18F-FDG, injected dose approximately 10 mCi Kinetic modeling:3-compartment model	AD BBB K1 values(mL/min/g): Frontal cortex= 0.118 ± 0.015 , Temporal cortex= 0.126 ± 0.012 , Occipital cortex= 0.159 ± 0.021 , Entire cortex= 0.141 ± 0.008 , White matter= 0.102 ± 0.015 .
Piert et al. (1996)	8 AD, 10 NCI	No provided.	Kinetic FDG-PET parameters: 15 scans over 60 mins Frame times:0.5 min: 4 scans, 1 min: 3 scans, 2.5 min: 2 scans, 5 min: 2 scans, 10 min: 4 scans. FOV 10 cm above canthomeatal line Arterial sampling: 2 min post-injection, rapidly Kinetic modeling:three-compartment, four-parameter model	AD BBB K1 values (mL/min/g): Frontal cortex= 0.082 ± 0.017 , Temporal cortex= 0.082 ± 0.017 , Parietal cortex= 0.067 ± 0.013 , Occipital cortex= 0.080 ± 0.023 , Thalamus: 0.119 ± 0.016 , Caudate nucleus= 0.106 ± 0.021 , Putamen= 0.114 ± 0.025 , Cerebellar cortex= 0.129 ± 0.027
Mosconi et al. (2007)	7 mild AD, 6 NCI.	PET scanner: Siemens ECAT-931.	Kinetic FDG-PET parameters: Dynamic FDG PET protocol lasted 60 min, Consisted of 22 frames: 4×30 s, 8×60 s 10×5 min, FOV: 10 cm axial, 20 cm transaxial, Resolution: 6.2 mm in-plane axial, 6.7 mm cross-slice, 6.75 mm interslice distance, Kinetic modeling:Three-compartment kinetic model	AD BBB K1 values(mL/min/g): Hippocampus= 0.08 ± 0.07 , Lateral temporal lobe= 0.09 ± 0.09 , Inferior parietal lobule= 0.10 ± 0.09 Thalamus= 0.13 ± 0.10 , Prefrontal cortex= 0.09 ± 0.08 , Occipital cortex= 0.13 ± 0.11 , Posterior cingulate cortex= 0.10 ± 0.09 , Amygdala= 0.09 ± 0.06 , Cerebellum= 0.12 ± 0.11 , Parahippocampal gyrus= 0.09 ± 0.08 , Pons= 0.12 ± 0.06 , Whole brain= 0.10 ± 0.01
Gejl et al. (2017)	14 liraglutide AD patients, 17 placebo AD patients, 6 NCI	PET scanner: GE Advance	Kinetic FDG-PET parameters: Scan duration: 45 min total. Frames: 23 total - 6 frames 30 s, 7 frames 1 min, 5 frames 2 min, 5 frames 5 min. Field-of-view: 15 cm. Spatial resolution: 4–5 mm at center. Simultaneous acquisition in 47 transaxial planes. Radiotracer administration: [18 F]-FDG, 200 MBq injected over 10 s. Kinetic modeling: 3-tissue compartment model	Subjects BBB glucose transfer capacity ($\mu\text{mol/g/min}$): Placebo group baseline= 0.72, Placebo group 6 months= 0.72 Liraglutide group baseline= 0.57, Liraglutide group 6 months= 0.91 Healthy controls= 1.02
[11 C]-verapamil PET van Assema et al. (2012)	13 mild AD, 14 NCI	PET Scanner: ECAT EXACT HR+	11C-verapami-PET parameters: Dynamic 3D scan for 60 min, 20 frames from 1 to 600 s Continuous arterial blood sampling at 5 mL/min for 5 min, 2.5 mL/min afterwards Additional manual blood samples at multiple timepoints Plasma input function from continuous curve, manual samples and metabolite correction Kinetic modeling:2 tissue compartment model	AD BBB 1C-verapami binding potential non-displaceable: Global= 2.18 ± 0.25 , Frontal= 2.12 ± 0.28 , Parietal= 2.15 ± 0.26 , Temporal= 2.29 ± 0.27 ,Occipital= 2.20 ± 0.25 , Posterior cingulate= 2.20 ± 0.35 , Anterior cingulate= 2.06 ± 0.35 , Medial temporal lobe= 2.97 ± 0.42 , Cerebellum= 1.99 ± 0.24
Deo et al. (2014)	9 mild AD, 9 NCI	PET Scanner: GE Advance PET	11C-verapami-PET parameters: Scan sequence: 16×5 s frames, 7×10 s frames, 5×30 s frames, 5×60 s frames, 5×120 s frames. Total scan time: 20 mins post-injection Blood samples: Arterial for 12 subjects, venous for rest Input function: Estimated from image data Injection: Intravenous bolus over 1 min Kinetic modeling:1-tissue compartment model to derive K1 maps	No specific values provided.

NCI = individual with no cognitive impairment,MCI = mild cognitive impairment, AD=Alzheimer's disease, BBB=Blood-Brain Barrier, APOE = Apolipoprotein E,

strategies. Current optimization efforts focus on substrates/inhibitors with attenuated P-gp affinity, exemplified by [11 C]-Metoclopramide and [18 F]-MC255. These novel agents enhance initial brain uptake kinetics, thereby enabling more robust quantification of P-gp functionality (Luurtsema et al., 2016; Mairinger et al., 2011). Beyond P-gp-targeted agents, radiotracer development has expanded to encompass other critical BBB components.] Emerging tracers probe diverse BBB properties including aquaporin dynamics, RAGE signaling, matrix metalloproteinase activity, and cell adhesion molecule expression.

AQP4-targeted probes like [11 C]TGN-020 exploit the unique distribution of aquaporins, AQP4 predominates at the BBB while AQP1 localizes to the blood-CSF barrier, providing spatially resolved functional assessments(Nakamura et al., 2011; Tomassoni et al., 2010). Parallel advancements are reshaping RAGE imaging paradigms. The development of [18 F]-RAGER and [18 F]-InRAGER tracers targeting distinct RAGE domains represents a significant innovation, though challenges persist regarding off-target binding to melatonin receptors(Drake et al., 2020). The translational potential of these techniques extends beyond static

BBB evaluation. Matrix metalloproteinase-PET demonstrates particular promise in differentiating early-stage BBB lesions with active leukocyte infiltration from quiescent pathological changes (Gerwien et al., 2016), potentially enabling cellular-level neuroinflammation monitoring. Similarly, vascular cell adhesion molecule-1 tracers initially validated in mouse atherosclerosis models (Bala et al., 2019) are being adapted for BBB integrity assessments, with intercellular adhesion molecule-1 analogues under active development. Collectively, these multimodal PET approaches illuminate BBB pathophysiology in AD with unprecedented resolution. Kinetic FDG-PET analyses reveal impaired glucose transport across the AD BBB, while [^{11}C]-verapamil-PET consistently documents P-gp dysfunction. Ongoing development of target-specific tracers continues to refine our understanding of transporter/protein alterations and their spatiotemporal relationship to AD progression. Through non-invasive molecular imaging, PET technology stands to revolutionize our comprehension of BBB contributions to AD pathogenesis.

5. Challenges and future directions in BBB Imaging

Recent advancements in neuroimaging and computational analytics have revolutionized the characterization of BBB dysfunction in AD. Nevertheless, critical technical and translational challenges persist in four key domains: multimodal data integration, molecular stratification, methodological standardization, and clinical implementation. Emerging artificial intelligence (AI)-driven frameworks demonstrate unprecedented potential to resolve inter-modal discrepancies through adaptive feature fusion and deep learning-enhanced resolution optimization across structural, functional, and metabolic imaging modalities, including DCE-MRI, water exchange MRI, and PET. Concurrently, the field confronts ethical-technological frontier challenges in clinical translation, particularly concerning gadolinium deposition risks and radiation exposure thresholds. These dual imperatives - technological innovation and biosafety optimization - necessitate systematic solutions to enable precision neurology applications while maintaining patient-centric ethical standards.

5.1. Multimodal BBB Imaging: Technical Barriers and AI-Driven Integration

The intricate architecture and functional heterogeneity of the BBB necessitate multimodal imaging approaches to comprehensively evaluate its spatiotemporal evolution during AD progression. However, critical challenges persist in reconciling discordant findings across modalities and translating these insights into clinically actionable biomarkers. A primary constraint arises from inherent trade-offs between spatial resolution, molecular specificity, and temporal dynamics across imaging techniques. While DCE-MRI remains the clinical gold standard for detecting BBB leakage, its application in AD research faces substantial methodological hurdles. The pathophysiological BBB disruption in AD manifests as minute leakage rates (0.01–0.1 mL/100 g/min) approaching DCE-MRI's detection threshold, where low signal-to-noise ratios frequently obscure the distinction between true leakage and background fluctuations (van de Haar et al., 2017). Technical confounders including partial volume effects from CSF contamination, scanner drift due to magnetic field inhomogeneity, and motion artifacts further compromise measurement accuracy, particularly in longitudinal investigations (Armitage et al., 2011; Chagnot et al., 2021). The lower detection sensitivity of gadolinium-based approaches is constrained by contrast agent molecular weight (938 Da) and relaxometric properties, requiring careful balance between pulse sequence parameters (e.g., flip angle, TR/TE), temporal sampling frequency, and spatial coverage (Elschot et al., 2021). Paradoxically, while enhanced temporal resolution improves pharmacokinetic modeling precision (e.g., K_{Trans} and V_e estimation), it inevitably sacrifices spatial resolution - a critical limitation given the region-specific nature of early BBB breakdown in AD-related neurovascular pathology (Larsson et al., 2013; Wardlaw

et al., 2009). Emerging water-exchange MRI techniques circumvent gadolinium limitations by exploiting endogenous water (18 Da) as a natural tracer, capitalizing on its bidirectional transport across transcellular (aquaporin-mediated) and paracellular (TJ-dependent) pathways (Dickie et al., 2020; Uchida et al., 2023). This molecular weight advantage confers heightened sensitivity to incipient BBB dysfunction, with preclinical models demonstrating water-exchange MRI's capability to detect subtle barrier alterations earlier than DCE-MRI (Dickie et al., 2019; Tiwari et al., 2016). Nevertheless, current spatial resolutions (>1 mm³) remain inferior to DCE-MRI's anatomical delineation capabilities (Montagne et al., 2022; Varatharaj et al., 2018). Water-exchange methodologies encompassing ASL and filtration exchange imaging probe BBB water dynamics - processes governed by both passive diffusion and active transport mechanisms (Dickie et al., 2020; Nilsson et al., 2013). However, conflicting correlations between K_w and AD biomarkers (positive associations with CSF A β 42 in some studies (Gold et al., 2021) versus negative correlations with A β 42/A β 40 ratios in others (Lin et al., 2021)) underscore the interpretive ambiguity introduced by the complex biophysics of water exchange. Unlike the direct permeability measures provided by gadolinium extravasation, water transport reflects composite contributions from blood flow, capillary permeability, and interstitial clearance (Bai et al., 2018; Springer et al., 2014). Advanced multicompartment kinetic modeling incorporating parameters for intravascular, extravascular extracellular, and intracellular water pools shows promise in disentangling specific BBB exchange signals from confounding factors (Dickie et al., 2020; Elschot et al., 2021). The evolution of metabolic BBB imaging is exemplified by DGE-MRI, which tracks D-glucose transport through GLUT1-mediated transcellular pathways and paracellular leakage routes (Patching, 2016). With molecular weight (180 Da) intermediate between water and gadolinium chelates, DGE-MRI demonstrates enhanced sensitivity to subtle barrier disruption while providing functional insights into neurovascular coupling. Clinical validation reveals superior lesion conspicuity in early AD compared to DCE-MRI, correlating with histopathological markers of BBB injury, neuroinflammation, and heightened metabolic activity (Xu et al., 2015b). Technical limitations persist, however, as the technique's reliance on CEST effects necessitates high-field systems to overcome signal attenuation at physiological glucose concentrations.

AI-driven multimodal data fusion will be achieved through developing adaptive feature extraction algorithms to integrate multidimensional data from DCE-MRI, water-exchange MRI, and DGE-MRI, effectively addressing cross-modal inconsistencies while generating spatiotemporally correlated biomarker systems to enhance early detection of AD-related BBB impairment (Rudroff et al., 2024; Sui et al., 2023). Concurrently, intelligent image enhancement frameworks will employ deep learning architectures to transcend conventional imaging limitations, enabling subthreshold leakage detection with preserved anatomical resolution through spatiotemporal dynamic modeling of water-exchange and metabolic transport processes, thereby significantly improving quantitative characterization of subtle BBB dysfunction (Kim et al., 2024; Lu et al., 2024). Furthermore, machine learning-optimized closed-loop imaging protocols will intelligently reconcile inherent trade-offs between spatial resolution, temporal sampling frequency, and molecular sensitivity, facilitating personalized scanning scheme design to ensure longitudinal monitoring reliability and reproducibility in BBB functional assessments (Contijoch et al., 2020).

5.2. Breaking molecular stratification challenges: fusion of PET-MRI biomarkers in BBB

The emerging concept of molecularly stratified BBB dysfunction introduces unprecedented complexity in cross-modal biomarker validation. This spatial-temporal heterogeneity manifests as distinct pathophysiological signatures across neurovascular units, where transporter-specific deficits (e.g., GLUT1-mediated glucose transport)

may coexist with preserved structural integrity in adjacent regions. PET has emerged as an indispensable molecular imaging modality, providing complementary insights to conventional MRI through its capacity for targeted interrogation of BBB pathophysiology. The technique's strength lies in its ability to quantify nanomolar-level alterations in specific transport systems - exemplified by [18 F]-FDG's detection of GLUT1-mediated glucose hypometabolism (Pierr et al., 1996) and [11 C]-verapamil's mapping of P-gp efflux dysfunction (van Assema et al., 2012). Paradoxically, while PET's spatial resolution (2–5 mm isotropic) lags behind MRI's submillimeter capabilities (0.5–1 mm) (Mairinger et al., 2011), its picomolar sensitivity enables absolute quantification of both transporter kinetics (e.g., P-gp efflux rate constants) and metabolic flux parameters (e.g., cerebral metabolic rate of glucose) through dynamic compartmental modeling (Barrio et al., 2020). Nevertheless, clinical implementation confronts multifaceted technical barriers. Tracer pharmacokinetics require meticulous optimization to balance target specificity and nonspecific binding - a critical consideration given verapamil's high P-gp affinity may mask early-stage transporter dysfunction through signal saturation effects (Mairinger et al., 2011). Furthermore, partial volume errors induced by PET's limited resolution disproportionately affect small structures like the hippocampus, confounding interpretation of region-specific BBB impairment (Langer et al., 2007). These limitations stand in stark contrast to emerging MRI-based techniques such as DGE-MRI, which leverages endogenous glucose as a CEST contrast agent for noninvasive GLUT1 assessment (Knutsson et al., 2018). However, DGE-MRI introduces its own methodological complexities, particularly in arterial input function quantification and susceptibility to systemic glycemic fluctuations that may obscure cerebral glucose dynamics. This methodological dichotomy underscores the imperative for integrative neuroimaging frameworks. A promising approach combines transporter-specific PET biomarkers with spatially resolved MRI metrics through advanced coregistration algorithms. For instance, correlating DGE-MRI's GLUT1 sensitivity with DCE-MRI leakage maps could disentangle metabolic deficits (reduced glucose uptake) from structural barrier breakdown (contrast extravasation) within the same neuroanatomical region (Xu et al., 2015b). Such multimodal paradigms not only enhance diagnostic specificity but also enable longitudinal tracking of BBB pathophysiology across its molecular-structural continuum. The dynamic evolution of BBB dysfunction introduces additional layers of complexity for modality-specific interpretation. Neuropathological studies have established that pericyte degeneration occurs early in AD pathogenesis, preceding A β plaque deposition by decades. This cellular loss disrupts low-density LRP1-mediated A β clearance mechanisms, creating a self-reinforcing cycle of impaired efflux and accelerated accumulation (Montagne et al., 2020). Advanced imaging techniques now reveal corresponding temporal hierarchies in vivo: DCE-MRI detects subtle hippocampal leakage up to 15 years before volumetric atrophy becomes apparent (Montagne et al., 2015), while ASL identifies impaired perivascular clearance in APOE4 carriers - a phenotype strongly associated with glymphatic dysfunction and AQP4 polarization defects (Joseph et al., 2020; Uchida et al., 2022). These observations suggest distinct temporal niches for different imaging modalities, with water-exchange MRI potentially capturing early functional declines in clearance pathways, whereas gadolinium-based techniques predominantly reflect later-stage structural damage to the neurovascular unit. Resolving these spatiotemporal complexities demands systematic longitudinal studies combining: (1) Regional Ktrans mapping for anatomical localization of gross barrier failure; (2) Kw quantification via water-exchange MRI for functional assessment of clearance pathways; (3) DGE-MRI for evaluating neurovascular metabolic coupling; (4) PET imaging of transporter activity (e.g., P-gp function using [11 C]-verapamil). Correlation of these multimodal datasets with histopathological staging and multi-omics profiles will enable mechanistic decoding of BBB breakdown trajectories.

Emerging AI methodologies present promising solutions to current

technical challenges through synergistic approaches. First, machine learning models trained on multiparametric datasets could implement dynamic physiological compensation by distinguishing systemic confounders (e.g., blood glucose fluctuations) from authentic cerebral glucose dynamics in CEST-MRI, which may significantly improve diagnostic precision for early-stage metabolic abnormalities (Owess et al., 2024). In addition, deep learning architectures could further enable multimodal fusion through automated cross-modal feature correlation analysis, effectively combining PET's molecular sensitivity with MRI's spatial precision to concurrently visualize transporter dysfunction and structural barrier integrity (Da-Ano et al., 2024; Li et al., 2024). Complementing these advancements, reinforcement learning systems may accelerate tracer optimization by engineering next-generation PET probes with balanced target affinity and nonspecific binding characteristics, thereby extending the detection window for early transporter dysfunction before signal saturation occurs (Wang et al., 2024). Deep learning architectures trained on multi-scale biological data could disentangle the relative contributions of endothelial TJ loss, basement membrane degradation, and pericyte dysfunction to observed imaging phenotypes (Saber-Bosari et al., 2020). Such integrative approaches may ultimately enable precision BBB phenotyping, transforming our ability to stratify AD patients based on distinct neurovascular dysfunction endotypes and to monitor targeted therapeutic interventions.

5.3. Methodological Harmonization: Protocol Standardization and Dynamic Biomarker Validation

Technical standardization persists as another critical barrier. The lack of standardized methodologies continues to impede cross-study comparability in BBB assessment. A critical issue arises from the variability in pharmacokinetic modeling approaches, where the selection between Patlak and extended Tofts models introduces substantial discrepancies in Ktrans quantification. These differences are exacerbated by inconsistent acquisition protocols, particularly variations in temporal resolution and contrast agent administration schedules (Barnes et al., 2016; Chagnot et al., 2021). Similarly, ASL-derived Kw measurements demonstrate concerning interstudy variability, primarily attributable to divergent magnetization preparation strategies across research groups (Gregori et al., 2013; Mahroo et al., 2021). This methodological heterogeneity not only undermines the reproducibility of multicenter studies but also complicates longitudinal monitoring of dynamic BBB alterations in neurodegenerative processes. The validation conundrum further compounds these technical challenges. Current practice relies on indirect biomarkers such as CSF albumin quotient and soluble platelet derived growth factor receptor- β as surrogate measures of BBB integrity (Cicognola et al., 2023; Erickson and Banks, 2019). However, these biomarkers suffer from dual limitations: they lack the spatial resolution to localize barrier defects and exhibit poor mechanistic specificity in distinguishing between endothelial dysfunction, pericyte degeneration, and other neurovascular unit pathologies. This fundamental gap in validation methodology creates uncertainty in correlating imaging biomarkers with underlying biological processes.

Recent advances propose complementary strategies to address these standardization issues. Initially, harmonized acquisition protocols have been established, exemplified by consensus guidelines establishing standardized DCE-MRI scan durations and gadolinium dosing regimens to improve inter-site comparability (Barnes et al., 2016). Concurrently, machine learning frameworks are being developed that synergistically integrate multiparametric MRI data with vascular risk profiles derived from plasma proteomics and transcriptomics, thereby enhancing the biological interpretability of imaging biomarkers (Lee et al., 2022; Yang et al., 2022). Notably, AI algorithms demonstrate the capability analyze multi-center historical datasets to automatically generate optimized imaging parameter combinations. This effectively minimizes technical heterogeneity across research institutions, consequently improving the biological relevance of cross-study comparisons for critical quantitative

parameters such as Ktrans and Kw (Foti and Longo, 2024; Zerunian et al., 2022). Significantly, time-series modeling-based AI systems enable the capture of nonlinear evolutionary patterns in BBB permeability dynamics. By identifying critical transition points during neurodegenerative disease progression, these systems establish standardized analytical pathways for longitudinal investigations. This innovation addresses the inherent limitations of conventional static assessment methodologies in monitoring dynamic pathological processes (Chen et al., 2017; Lin et al., 2024).

5.4. Clinical translation: navigating ethical barriers and technological frontiers

The clinical implementation of BBB imaging biomarkers confronts multifaceted challenges encompassing bioethical considerations and technological limitations. Of particular concern is the cumulative gadolinium deposition associated with repeated contrast-enhanced MRI examinations, which poses potential neurotoxic risks in elderly populations with compromised renal clearance mechanisms (Montagne et al., 2019). Concurrently, the ionizing radiation exposure inherent to PET imaging imposes critical constraints on longitudinal monitoring paradigms, particularly in vulnerable patient cohorts. Emerging technological innovations present promising avenues to circumvent these limitations: ultrahigh-field MRI systems demonstrate enhanced sensitivity for detecting subtle BBB leakage through improved signal-to-noise ratios and spatial resolution, while next-generation PET radiotracers targeting specific molecular pathways (e.g., receptor for RAGE and MMP-9) offer improved target specificity with reduced radiation dosages (Drake et al., 2020; Gerwien et al., 2016). However, the clinical adoption of these advanced modalities remains hindered by substantial infrastructure costs, limited availability of specialized imaging centers, and the need for standardized acquisition protocols. To realize the transformative potential of BBB imaging in precision neurology, concerted efforts must focus on developing cost-effective, non-invasive biomarkers compatible with routine clinical workflows. Priority research directions should include: (1) optimization of exogenous contrast agent-independent imaging techniques for quantitative BBB assessment, (2) AI-powered low-dose imaging reconstruction techniques that enhance image quality through deep learning while maintaining diagnostic sensitivity with reduced PET radiation exposure, and (3) establishment of international registries to facilitate multicenter validation of novel biomarkers. Such advancements will be essential for enabling population-level screening of preclinical BBB dysfunction and objective evaluation of emerging neurotherapeutics.

6. Conclusion

This work systematically reviews the pathological mechanisms of BBB disruption in AD and the latest advancements in in vivo imaging techniques. As a core component of the neurovascular unit, the BBB maintains cerebral homeostasis by regulating A β clearance. Its dysfunction is a key driver of early AD pathology. This work provides a detailed analysis of the cellular and molecular structure of the BBB, emphasizing the synergistic effects of A β transport imbalance and neuroinflammation in AD. In terms of imaging technologies, DCE-MRI quantifies Ktrans using gadolinium contrast agents, while water exchange MRI noninvasively assesses BBB permeability. DGE-MRI reveals GLUT1-mediated metabolic defects. PET technology, utilizing tracers such as [18 F]-FDG and [11 C]-verapamil, targets glucose metabolism and P-gp efflux function, offering unique perspectives for molecular-specific imaging. However, existing techniques face challenges, including insufficient sensitivity, model dependence, and difficulties in integrating multimodal data. Future directions require combining multimodal integration, ultrahigh-field systems, and artificial intelligence-driven analytics to decode region-specific BBB pathology. This strategy aims to precisely identify early AD BBB phenotypes and

guide personalized therapies. Advancements in this field will deepen our understanding of AD neurovascular pathology and promote a paradigm shift in therapeutic strategies from solely targeting amyloid proteins to multi-target interventions.

Author contributions

Guarantors of integrity of entire study, **Xiaohe Tian, Qiyong Gong**; study concepts/study design or data acquisition or data analysis/interpretation, **all authors**; manuscript drafting, **Haolin Yin, Yuepeng Deng**; drew figures and organized the tables for this paper, **Haolin Yin, Zihao Lu**; Manuscript revision for important intellectual content, **all authors**; approval of final version of submitted manuscript, **all authors**; agrees to ensure any questions related to the work are appropriately resolved, **all authors**.

Consent for publication

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Ethics approval and consent to participate

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Declaration of Competing Interest

All authors declare no competing interests.

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Data Availability

Not applicable

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